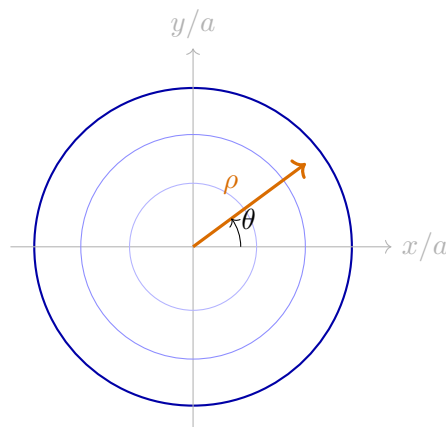


SELF-STUDY OPTICS SERIES

Zernike Coefficients

for Mirrors and Lenses

From first principles to wavefront analysis



Detailed derivations, worked examples, computational checks,
and exercises with solutions

10 September 2026

How to study this book

A Zernike coefficient is a number describing one component of a surface or a wavefront over a specified pupil. It is not a universal constant belonging to a piece of glass or metal. This book develops the mathematics needed to calculate those numbers and the optics needed to interpret them correctly.

You need no previous optics course. Part I introduces units, elementary calculus, optical path length, reflection, refraction, image formation, pupils, and a reproducible ray-tracing procedure. Work through these chapters with pencil and paper. The mathematical explanations introduce the operations as they become useful; if differentiation and integration are new to you, repeat the short examples before moving on.

Part II builds the Zernike basis rather than presenting it as a table to memorize. It then connects the coefficients to mirror figure, lens thickness, reference focus, image quality, and measured data. The examples are specified teaching designs, not specifications for an existing commercial instrument. Every quoted numerical coefficient depends on the stated units, aperture, wavelength when used, and reference.

A practical study sequence. Use approximately two sessions for the mathematical tools, three for rays and first-order imaging, two for pupils and optical path, four for orthogonality and coefficient extraction, and four or more for mirror/lens examples and numerical fitting. A session is complete when you can rederive its principal equation without copying the text. These are suggested blocks of work, not a promise that research proficiency takes a fixed number of hours.

Companion material. The companion book in `../abberations/` develops field-dependent aberration theory and the distinction between third-, fifth-, and higher-order ray aberrations. Both books contain the same introductory foundations so either can be studied independently. The file `../verify_optics.py` reproduces selected mathematical and ray-tracing checks; `../numerical_checks.txt` records its output. Detailed exercises and solutions in this volume turn the derivations into calculations you can adapt.

Research scope. The aim is to prepare you to begin supervised design, simulation, and metrology work: define a prescription, compute a wavefront, fit modes, test convergence, and interpret residuals. Vector diffraction, electromagnetic multilayer design, inverse-problem regularization, fabrication process modelling, and full system optimization require further study. Wherever a simplified model is introduced, its assumptions are stated before it is used.

Notation and convention card

Keep this page beside your calculations. Individual examples may introduce local symbols, but each defines its geometry before using it.

Symbol	Meaning	Typical units or range
λ_0	Vacuum wavelength	nm, μm , or mm; convert consistently
n, n', n_g	Refractive indices in incident, output, and glass regions	Dimensionless
$\mathcal{L}$	Optical path length $\int n d\ell$	Length
S	Eikonal; phase expressed as optical length	Length
W	Actual minus ideal eikonal at the stated comparison points	Length; waves only when explicitly divided by λ_0
ϕ	Optical phase, $2\pi S/\lambda_0$	Radians
$\mathbf{d}, \mathbf{N}$	Unit ray direction and unit surface normal	Dimensionless vectors
$R, c = 1/R$	Vertex radius and curvature	Length and inverse length
K	Conic constant in the specified sag equation	Dimensionless
a, r	Physical pupil radius and physical radial coordinate	Length
ρ, θ	Normalized radius r/a and pupil azimuth	$0 \leq \rho \leq 1$; radians
u, v	Often normalized Cartesian pupil coordinates in wavefront sections	Dimensionless; local definitions take precedence
$u, p = nu$	In first-order ray matrices, small angle and reduced angle	Radians and index times radians
H	Normalized field magnitude in invariant expansions	Dimensionless when normalized
f, L	Focal length; distance from comparison plane to reference focus	Length; generally different
Z_n^m	Real RMS-normalized disk mode	Dimensionless; here n is a radial integer, not an index of refraction
c_n^m	Coefficient multiplying Z_n^m in a wavefront expansion	Same units as W
$\langle \cdot \rangle$	Specified pupil average; uniform area by default	Preserves the units of its argument
σ_W	RMS wavefront deviation after the stated removals	Length
$\Delta \mathbf{x}$	Actual minus ideal transverse ray intercept	Length

Geometry and OPD. For a transmitting system the local forward axis is normally $+z$. A signed spherical radius is positive when its centre of curvature is downstream. Mirror examples explicitly specify the direction before and after reflection. Our OPD sign is $W = S_{\text{actual}} - S_{\text{ideal}}$, giving the leading forward image-space relation $\Delta \mathbf{x} \simeq (L/n') \nabla_{\perp} W$. Changing the definition of OPD reverses the corresponding signed coefficients.

Modes and pupil. Unless explicitly stated otherwise, Zernike functions are real, orthonormal under the full-disk area average, with $m > 0$ for $\cos(m\theta)$ and $m < 0$ for $\sin(|m|\theta)$. A pair (n, m) identifies a function; a single number requires a named ordering convention. Coefficients depend on the pupil radius, coordinate surface, obscurations or weights, reference sphere, and removed modes.

Order and meaning. A wavefront polynomial's total degree in pupil and field variables, the degree of its transverse ray error, and the radial index of a Zernike mode answer different questions. Terms called “spherical” or “coma” in a Zernike fit include lower powers needed for orthogonality. They are not simply the corresponding highest monomials. Surface Zernike amplitudes are not wavefront amplitudes until the reflection or refraction conversion and pupil mapping have been applied.

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Part I

Foundations: from light to an optical model

Chapter 1

The mathematical language of optics

1.1 What an optical model describes

Imagine a small luminous point, a lens, and a sheet of paper. Light leaving the point travels in many directions. Some of those directions reach the lens. A successful imaging system redirects that light into a small region on the paper. A mirror accomplishes redirection by reflection; a lens does it by refraction. An optical design specifies the surfaces, their positions, the materials between them, and which rays are admitted.

There are two complementary descriptions. A *ray* describes a local direction of energy propagation in the geometrical approximation. A *wavefront* describes equal phase. Rays let us calculate intersections and image positions. Wavefronts tell us whether light arriving by different paths adds constructively. An aberration can therefore be described by misplaced rays or by an error in optical phase. The two descriptions are connected but their numerical measures have different units.

Before deriving an equation, write down what its variables represent. A surface height, a ray height, and an optical path difference may all be measured in millimetres; they are nevertheless different quantities. Similarly, a physical pupil radius and a normalized radius may have the same numerical value in one example but different dimensions.

1.2 Units, angles, and dimensionless coordinates

We use the vacuum wavelength λ_0 . The units most often encountered are

$$1 \text{ mm} = 10^{-3} \text{ m}, \quad 1 \text{ }\mu\text{m} = 10^{-6} \text{ m}, \quad 1 \text{ nm} = 10^{-9} \text{ m}.$$

Thus $633 \text{ nm} = 0.000633 \text{ mm}$. If an optical path error is $W = 31.65 \text{ nm}$, its value in waves at this wavelength is $W/\lambda_0 = 0.05$. The phase error is $2\pi W/\lambda_0 = 0.31416$ radians. At 13.5 nm the same path error is about 2.344 waves. A surface acceptable at one wavelength may be unacceptable at another even when the ray geometry is identical.

An angle in radians is arc length divided by radius. One full revolution is 2π radians, so $1^\circ = \pi/180$ radians. Series such as $\sin u \simeq u$ assume radians. Substituting $u = 5$ for five degrees would be a serious error; the correct number is approximately 0.08727.

For a circular pupil of physical radius a , define

$$x = a\rho \cos \theta, \quad y = a\rho \sin \theta, \quad 0 \leq \rho \leq 1, \quad 0 \leq \theta < 2\pi. \quad (1.1)$$

Here x, y, a have units of length and ρ is dimensionless. The angle θ increases counterclockwise from the positive x axis toward the positive y axis when looking at the plotted pupil. State the

physical viewing direction separately when comparing two instruments, because looking from the opposite side reverses handedness.

1.3 Functions, slopes, and curvature

A function assigns an output to an input. A sag function $z = s(x, y)$ gives a surface's axial position at transverse point (x, y) . For a rotationally symmetric surface, s depends only on $r = \sqrt{x^2 + y^2}$.

The derivative is a limiting slope:

$$\frac{ds}{dr} = \lim_{\Delta r \rightarrow 0} \frac{s(r + \Delta r) - s(r)}{\Delta r}.$$

For $s(r) = cr^2/2$, expand $(r + \Delta r)^2$, subtract r^2 , and divide by Δr to get $cr + c\Delta r/2$. Taking the limit gives $s'(r) = cr$. The second derivative $s''(r) = c$ describes how fast the slope changes. For a shallow spherical surface, $c = 1/R$ is its vertex curvature.

A partial derivative varies one coordinate while holding the other fixed. The transverse gradient is

$$\nabla_{\perp} W = \left(\frac{\partial W}{\partial x}, \frac{\partial W}{\partial y} \right).$$

For $W = A(x^2 + y^2)^2$,

$$\frac{\partial W}{\partial x} = 4Ax(x^2 + y^2), \quad \frac{\partial W}{\partial y} = 4Ay(x^2 + y^2).$$

The coefficient A here has units of inverse length cubed if W is a length. If instead $W = B\rho^4$, then B has units of length. The two coefficients satisfy $B = Aa^4$. Confusing them gives errors that grow as the fourth power of the aperture radius.

The chain rule handles normalization. With $u = x/a$ and $v = y/a$,

$$\frac{\partial W}{\partial x} = \frac{1}{a} \frac{\partial W}{\partial u}.$$

The factor $1/a$ is essential when converting a fitted wavefront into a physical ray angle.

1.4 Integrals and the area measure on a disk

An integral adds infinitesimal contributions. The one-dimensional identity

$$\int_0^1 \rho^q d\rho = \frac{1}{q+1} \quad (q > -1)$$

follows because the derivative of $\rho^{q+1}/(q+1)$ is ρ^q .

A small polar patch has radial width dr and tangential width $r d\theta$. Its area is therefore $r dr d\theta$. After setting $r = a\rho$,

$$dA = a^2 \rho d\rho d\theta.$$

The area of the disk follows immediately: $a^2 \int_0^{2\pi} \int_0^1 \rho d\rho d\theta = \pi a^2$. The area average of a function is thus

$$\langle F \rangle = \frac{1}{\pi} \int_0^{2\pi} \int_0^1 F(\rho, \theta) \rho d\rho d\theta. \quad (1.2)$$

In particular,

$$\langle \rho^q \rangle = \frac{2}{q+2}, \quad \langle \rho^2 \rangle = \frac{1}{2}, \quad \langle \rho^4 \rangle = \frac{1}{3}, \quad \langle \rho^6 \rangle = \frac{1}{4}.$$

These simple moments will repeatedly replace lengthy integrations.

The mean-subtracted root-mean-square value of a wavefront is

$$\sigma_W = \sqrt{\langle (W - \langle W \rangle)^2 \rangle} = \sqrt{\langle W^2 \rangle - \langle W \rangle^2}. \quad (1.3)$$

To obtain the second equality, expand the square, use linearity of the integral, and recognize that $\langle W \rangle$ is a constant. For $W = B\rho^2$, the mean is $B/2$ and the variance is $B^2(1/3 - 1/4) = B^2/12$, giving $\sigma_W = |B|/\sqrt{12}$. The peak-to-valley error is $|B|$, a different statistic.

1.5 Taylor expansions and what an omitted term means

Near $\epsilon = 0$, the binomial expansion gives

$$(1 + \epsilon)^{1/2} = 1 + \frac{\epsilon}{2} - \frac{\epsilon^2}{8} + \frac{\epsilon^3}{16} - \frac{5\epsilon^4}{128} + O(\epsilon^5), \quad (1.4)$$

$$(1 - \epsilon)^{1/2} = 1 - \frac{\epsilon}{2} - \frac{\epsilon^2}{8} - \frac{\epsilon^3}{16} - \frac{5\epsilon^4}{128} + O(\epsilon^5). \quad (1.5)$$

The notation $O(\epsilon^5)$ means that the neglected remainder scales no faster than a constant times $|\epsilon|^5$ sufficiently near zero. It is not an error estimate valid at arbitrary aperture. To use an expansion reliably, identify its dimensionless small quantity and check convergence at the largest ray height of interest.

For example,

$$\sqrt{L^2 + r^2} = L\sqrt{1 + r^2/L^2} \quad (1.6)$$

$$= L + \frac{r^2}{2L} - \frac{r^4}{8L^3} + \frac{r^6}{16L^5} + O(r^8/L^7), \quad L > 0. \quad (1.7)$$

Every term has dimensions of length. The expansion parameter is $(r/L)^2$. For $r/L = 0.05$, the quartic correction is smaller than the quadratic term by $(r/L)^2/4 = 0.000625$. That may be tiny geometrically but still many nanometres of optical path, which can matter in a short-wavelength instrument.

For a sphere with positive radius R and vertex at $z = 0$,

$$s(r) = R - \sqrt{R^2 - r^2} \quad (1.8)$$

$$= \frac{r^2}{2R} + \frac{r^4}{8R^3} + \frac{r^6}{16R^5} + \frac{5r^8}{128R^7} + \dots \quad (1.9)$$

The first term is a parabola in a meridional section. A sphere and a paraboloid can therefore have the same vertex curvature while differing away from the axis. Those apparently small higher-power differences are a central source of aberration.

1.6 Vectors and a minimal introduction to matrices

A three-dimensional vector $\mathbf{r} = (x, y, z)$ describes position or displacement. Its length is $|\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$. The dot product

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z = |\mathbf{a}| |\mathbf{b}| \cos \gamma$$

measures the component of one vector along the other; γ is their included angle. A unit vector has length one. A straight ray is $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{d}$, with unit direction $\mathbf{d}$ and distance parameter t in units of length.

A matrix encodes a linear operation. For example,

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} y \\ p \end{pmatrix} = \begin{pmatrix} Ay + Bp \\ Cy + Dp \end{pmatrix}.$$

If operation M_1 acts first and M_2 acts second, the combined result is $M_2M_1\mathbf{v}$. The rightmost matrix acts first. Changing that order usually changes the answer because refraction followed by translation is physically different from translation followed by refraction.

Chapter 2

Why light reflects, refracts, and accumulates phase

2.1 The domain of ray optics

Light is an electromagnetic wave. In a simple isotropic transparent medium the refractive index n relates the phase speed to the vacuum speed by $v = c/n$. In this introductory model, n is real and may depend on wavelength. Absorption and multilayer reflection require a complex optical response, discussed later where relevant.

Ray optics is a short-wavelength approximation: the relevant surface and field variations are smooth on the wavelength scale, and diffraction is not needed to locate the leading geometrical paths. The condition is not merely that a lens is bigger than a wavelength. A small sharp feature, an aperture edge, a caustic, or a narrow focal region can require a wave treatment even in a large apparatus. We use rays for the path geometry and wavefront phase for the first connection to image quality.

2.2 Optical path length and phase

If light travels a small geometric distance $d\ell$ in index n , it accumulates phase $2\pi n d\ell/\lambda_0$. Summing along a path gives

$$\mathcal{L} = \int n d\ell, \quad \phi = \frac{2\pi}{\lambda_0} \mathcal{L} + \phi_{\text{initial}}. \quad (2.1)$$

The quantity $\mathcal{L}$ is optical path length (OPL), not geometrical distance. In homogeneous segments it reduces to $\mathcal{L} = \sum_j n_j \ell_j$.

For 10 mm in air approximated by $n = 1$ followed by 5 mm of index 1.5, the geometrical length is 15 mm but the OPL is 17.5 mm. Replacing 5 mm of air by that glass increases OPL by 2.5 mm. Optical phase is controlled by that difference, not by the total thickness alone.

For two coherent scalar fields of real amplitudes A_1, A_2 , the intensity after superposition is proportional to

$$|A_1 e^{i\phi_1} + A_2 e^{i\phi_2}|^2 = A_1^2 + A_2^2 + 2A_1 A_2 \cos(\phi_1 - \phi_2).$$

This identity follows by multiplying the complex sum by its complex conjugate. Equal phases reinforce; a phase difference of π can cancel equal amplitudes. A geometrically small path mismatch can therefore have a large effect on image intensity.

2.3 Fermat's principle: stationary optical path

Fermat's principle states that the physical ray makes the optical path stationary under small allowed changes of path with endpoints held fixed. *Stationary* means that the first-order change vanishes. The path need not be a global minimum.

Derive Snell's law in a two-dimensional plane. Place a flat interface at $z = 0$, with source $A = (x_A, -h_1)$ in index n_1 and destination $B = (x_B, h_2)$ in index n_2 . Let the unknown crossing point be $P = (x, 0)$. The optical path is

$$\mathcal{L}(x) = n_1 \sqrt{(x - x_A)^2 + h_1^2} + n_2 \sqrt{(x_B - x)^2 + h_2^2}.$$

Differentiation gives

$$\frac{d\mathcal{L}}{dx} = n_1 \frac{x - x_A}{\sqrt{(x - x_A)^2 + h_1^2}} - n_2 \frac{x_B - x}{\sqrt{(x_B - x)^2 + h_2^2}}.$$

Each ratio is the sine of the corresponding signed angle to the interface normal. Setting the derivative to zero yields

$$n_1 \sin i = n_2 \sin t. \quad (2.2)$$

If $n_2 > n_1$ and the incident angle is positive, then $t < i$: the ray bends toward the normal. At normal incidence both angles vanish and the direction does not change, although phase accumulation does.

For reflection, put both endpoints on the same side of the interface and use the same index for both path segments. The analogous derivative sets the two signed tangential projections equal. The normal component reverses, yielding equality of incident and reflected angles measured from the normal. An ideal plane mirror redirects a converging or diverging bundle without acquiring optical power in its own local plane.

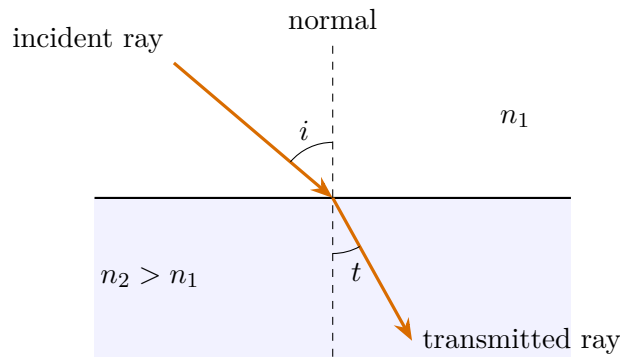


Figure 2.1: Angles in Snell's law are measured from the normal, not from the interface. Geometry is schematic.

2.4 Exact vector reflection

Let $\mathbf{d}$ be the incident unit direction and $\mathbf{N}$ any unit normal to the reflecting surface. Split $\mathbf{d}$ into normal and tangent components:

$$\mathbf{d}_\perp = (\mathbf{d} \cdot \mathbf{N})\mathbf{N}, \quad \mathbf{d}_\parallel = \mathbf{d} - \mathbf{d}_\perp.$$

Specular reflection preserves the tangent component and reverses the normal component. Therefore

$$\boxed{\mathbf{d}_r = \mathbf{d} - 2(\mathbf{d} \cdot \mathbf{N})\mathbf{N}.} \quad (2.3)$$

Replacing $\mathbf{N}$ by $-\mathbf{N}$ does not change this expression. With $\mathbf{d} = (0, 0, -1)$ and $\mathbf{N} = (0, 0, 1)$, reflection gives $(0, 0, 1)$ as expected. Reflection at a perfect plane changes direction; a spatially constant coating phase does not change the outgoing wavefront error after piston removal.

2.5 Exact vector refraction and total internal reflection

For refraction choose $\mathbf{N}$ to point into the incident medium, so $\mathbf{d} \cdot \mathbf{N} < 0$. Set

$$\eta = \frac{n_1}{n_2}, \quad c_i = -\mathbf{d} \cdot \mathbf{N}.$$

The incident tangent component is $\mathbf{d} + c_i \mathbf{N}$. Snell's law scales it by η , while normalization fixes the transmitted normal component. Thus

$$c_t = \sqrt{1 - \eta^2(1 - c_i^2)}, \quad (2.4)$$

$$\mathbf{d}_t = \boxed{\eta \mathbf{d} + (\eta c_i - c_t) \mathbf{N}}. \quad (2.5)$$

The negative normal component $-c_t \mathbf{N}$ sends the ray into the second medium. Verify normal incidence: $\mathbf{d} = -\mathbf{N}$ gives $\mathbf{d}_t = -\mathbf{N}$.

If the expression under the square root is negative, there is no propagating transmitted ray in this geometrical model. This is total internal reflection. For $n_1 > n_2$, its threshold is $\sin i_c = n_2/n_1$. A glass-to-air interface with $n_1 = 1.5$ has $i_c \simeq 41.81^\circ$. This is a propagation condition, not a statement that polarization-dependent phase can be neglected during reflection.

2.6 Eikonal, wavefront normals, and the ray limit

Write a monochromatic scalar field locally as

$$U(\mathbf{r}) = A(\mathbf{r})e^{ik_0 S(\mathbf{r})}, \quad k_0 = 2\pi/\lambda_0.$$

Here S has units of length and is called the eikonal. The suppressed time dependence is $e^{-i\omega t}$, so increasing spatial phase corresponds to propagation along ∇S . Substitute this form into the scalar Helmholtz equation $\nabla^2 U + k_0^2 n^2 U = 0$. Applying the product rule twice and cancelling the exponential gives

$$\nabla^2 A + 2ik_0 \nabla A \cdot \nabla S + ik_0 A \nabla^2 S - k_0^2 A |\nabla S|^2 + k_0^2 n^2 A = 0.$$

In the short-wavelength limit the terms proportional to k_0^2 dominate if the amplitude and material vary slowly. Their cancellation requires

$$|\nabla S|^2 = n^2, \quad \mathbf{d} = \frac{\nabla S}{n}. \quad (2.6)$$

A wavefront is a level surface $S = \text{constant}$, so its normal is ∇S . This explains the statement that rays are normal to wavefronts in an isotropic medium. At a caustic, several ray branches can reach the same point; a single smooth eikonal then requires branchwise interpretation. The full electromagnetic problem also contains vector and polarization effects absent from this scalar derivation.

Chapter 3

First-order images: mirrors, lenses, and ray matrices

3.1 Coordinates and the paraxial approximation

For transmitted systems, take the optical axis along $+z$, let ray height in a meridional section be y , and let u be its small signed angle to $+z$. The paraxial approximation uses

$$\sin u \simeq u, \quad \tan u \simeq u, \quad \cos u \simeq 1.$$

It retains only terms linear in ray heights and angles. It predicts ideal first-order imaging but discards the nonlinear terms that cause geometrical aberrations. “Paraxial” is an approximation to a physical ray, not a separate type of light.

A radius of curvature is positive if the centre of curvature lies downstream of its vertex. Thus the first surface of a usual biconvex lens has $R_1 > 0$ and the second has $R_2 < 0$. For a mirror we state the incident and reflected axes explicitly; blindly carrying the transmitted-system signs through a reversal of propagation is a frequent source of mistakes.

3.2 Paraxial power of one refracting surface

Near a vertex, $s(y) \simeq y^2/(2R)$ and the slope is y/R . A normal oriented approximately along $+z$ has small angle $\alpha \simeq -y/R$ to the axis. The signed incident and refracted angles relative to that normal are $u - \alpha$ and $u' - \alpha$. Linearizing Snell’s law gives

$$n(u - \alpha) = n'(u' - \alpha).$$

Rearrange and substitute $\alpha = -y/R$:

$$n'u' = nu - \frac{n' - n}{R}y. \quad (3.1)$$

Define reduced angle $p = nu$ and surface power $\Phi = (n' - n)/R$. Then $p' = p - \Phi y$. Positive power bends an initially axial ray at positive height toward the axis.

Suppose an on-axis real object is a positive distance s to the left of the vertex and its real image is a positive distance s' to the right. At the surface $u \simeq y/s$ and $u' \simeq -y/s'$. Substitution gives

$$\boxed{\frac{n}{s} + \frac{n'}{s'} = \frac{n' - n}{R}}. \quad (3.2)$$

This equation uses positive *real conjugate distances*, while R uses the downstream-centre convention. A virtual object or image gives a negative corresponding conjugate distance. Defining the convention first is more reliable than identifying a formula by its pattern of plus and minus signs.

3.3 The thin-lens equation derived twice

A thin lens has two surface powers whose separation is neglected. In surrounding air approximated as index one, add Eq. (3.1) at the two interfaces:

$$u_{\text{out}} = u_{\text{in}} - \Phi y, \quad \Phi = (n_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right).$$

For a collimated incident ray, $u_{\text{in}} = 0$. After distance f the ray has height $y + fu_{\text{out}} = y(1 - f\Phi)$. It reaches the axis when

$$\boxed{\frac{1}{f} = (n_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)}. \quad (3.3)$$

For an on-axis object at positive real distance s , the incoming angle is y/s . To reach an image at distance s' , the outgoing angle must be $-y/s'$. Therefore

$$\boxed{\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}}. \quad (3.4)$$

There is a wavefront derivation of the same result. Between fixed planes on opposite sides of a thin phase element, extra glass thickness $t(y) - t(0)$ contributes phase-equivalent path $(n_g - 1)[t(y) - t(0)]$. To quadratic order the biconvex thickness is

$$t(y) - t(0) \simeq \frac{y^2}{2} \left(\frac{1}{R_2} - \frac{1}{R_1} \right),$$

so the element contributes $-y^2/(2f)$. The path from the object contributes $y^2/(2s)$ and the path to the image contributes $y^2/(2s')$. For all small pupil rays to arrive with the same phase, their summed coefficient must vanish:

$$\frac{y^2}{2} \left(\frac{1}{s} + \frac{1}{s'} - \frac{1}{f} \right) = 0.$$

This reproduces the thin-lens equation. Extending a thickness-only model to fourth and sixth powers is a *phase-screen model*; it is not automatically the exact aberration of a finite lens, because ray bending inside the glass changes ray heights and optical paths.

Worked example. Let $n_g = 1.5$, $R_1 = +50$ mm, and $R_2 = -50$ mm. Equation (3.3) gives $1/f = 0.02 \text{ mm}^{-1}$ and $f = 50$ mm. An object at $s = 200$ mm gives $1/s' = 0.02 - 0.005 = 0.015 \text{ mm}^{-1}$, so $s' = 66.667$ mm. A central undeviated paraxial ray from an object height y_o gives image height $y_i = -s'y_o/s$. The lateral magnification is therefore $m = -1/3$; the negative sign means inversion.

3.4 A concave spherical mirror and the origin of its focal length

Choose the mirror vertex at $z = 0$, incident light travelling in $-z$, and a concave bowl $z = s(r) = R - \sqrt{R^2 - r^2}$ opening toward $+z$. For small r , $s'(r) \simeq r/R$. In a meridional section a unit

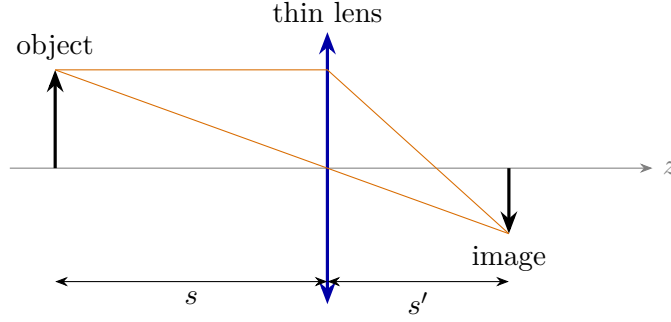


Figure 3.1: Paraxial imaging of an off-axis point. The drawing illustrates the construction; the numerical conjugates in the worked example differ from the schematic scale.

normal is approximately $(-r/R, 0, 1)$. Apply Eq. (2.3) to $\mathbf{d} = (0, 0, -1)$: the outgoing direction is approximately $(-2r/R, 0, 1)$. At distance f along $+z$, the transverse height is approximately $r - 2fr/R$. Hence

$$\boxed{f = R/2.} \quad (3.5)$$

For finite real object and image distances $s, s' > 0$ in front of this mirror, the geometric path through surface point r is

$$\sqrt{r^2 + [s - s(r)]^2} + \sqrt{r^2 + [s' - s(r)]^2}.$$

To quadratic order this is

$$s + s' + r^2 \left(\frac{1}{2s} + \frac{1}{2s'} - \frac{1}{R} \right) + O(r^4).$$

Stationarity for all paraxial heights requires $1/s + 1/s' = 2/R = 1/f$. The omitted quartic terms explain why a spherical mirror's finite-aperture rays generally fail to meet at one point. A paraboloid is exactly stigmatic for an on-axis collimated input under ideal specular geometric reflection; it is not generally perfect for other conjugates or off-axis fields.

3.5 Translation, refraction, and a thick lens

Use ray vector $(y, p)^\top$ with $p = nu$. Translation by physical distance d in constant index n gives $y' = y + du = y + (d/n)p$ and preserves p . Refraction preserves the surface intersection height and changes p by $-\Phi y$. The matrices are therefore

$$T(d, n) = \begin{pmatrix} 1 & d/n \\ 0 & 1 \end{pmatrix}, \quad S(\Phi) = \begin{pmatrix} 1 & 0 \\ -\Phi & 1 \end{pmatrix}. \quad (3.6)$$

For a lens of centre thickness d , internal index n_g , and surface powers Φ_1, Φ_2 ,

$$M = S(\Phi_2)T(d, n_g)S(\Phi_1) \quad (3.7)$$

$$= \begin{pmatrix} 1 - d\Phi_1/n_g & d/n_g \\ -\Phi_1 - \Phi_2 + d\Phi_1\Phi_2/n_g & 1 - d\Phi_2/n_g \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \quad (3.8)$$

The effective power is $\Phi_{\text{eff}} = -C = \Phi_1 + \Phi_2 - d\Phi_1\Phi_2/n_g$. In output air the effective focal length is $f_{\text{eff}} = 1/\Phi_{\text{eff}}$. A collimated input $(y_0, 0)$ emerges as (Ay_0, Cy_0) . Translation by a distance b to the axis requires $Ay_0 + bCy_0 = 0$, giving the back focal length measured from the second vertex:

$$\boxed{\text{BFL} = -A/C.} \quad (3.9)$$

Effective focal length is measured from a principal plane, an equivalent reference plane of the first-order system. Back focal length is measured from the last physical vertex. They are equal for an ideal thin lens in air but not for a general thick lens.

Worked thick-lens calculation. Retain $R_1 = +50$ mm, $R_2 = -50$ mm and $n_g = 1.5$, but use $d = 5$ mm. Then $\Phi_1 = \Phi_2 = 0.01$ mm⁻¹ and

$$A = D = \frac{29}{30}, \quad B = \frac{10}{3} \text{ mm}, \quad C = -\frac{59}{3000} \text{ mm}^{-1}.$$

Consequently,

$$f_{\text{eff}} = \frac{3000}{59} = 50.84746 \text{ mm}, \quad \text{BFL} = \frac{2900}{59} = 49.15254 \text{ mm}.$$

The paraxial focus is at global $z = 5 + 49.15254 = 54.15254$ mm if the first vertex is $z = 0$. A trace reporting a different finite-height axis crossing need not be wrong: spherical aberration shifts nonparaxial rays. Compare the trace as its entrance height tends to zero before diagnosing a discrepancy.

3.6 An imaging condition and a conservation check

Include all translations from object plane to image plane in a total matrix M . Then $y_i = Ay_o + Bp_o$. Rays from the same object point have fixed y_o but different p_o . They form a paraxial image when $B = 0$, making $y_i = Ay_o$ independent of the launch direction.

Both matrices in Eq. (3.6) have determinant one, so their product satisfies $AD - BC = 1$. For two paraxial rays with coordinates (y_1, p_1) and (y_2, p_2) , the combination $y_1 p_2 - y_2 p_1$ is unchanged by such a matrix. This follows by substitution and cancellation, leaving a factor $AD - BC$. This invariant is a useful error check and is related to the restrictions on trading image size against angular spread.

Chapter 4

Pupils, reference wavefronts, and the meaning of an error

4.1 Aperture stop, entrance pupil, and exit pupil

The *aperture stop* is the element that limits the angular cone accepted from an axial object point. Its opening may be a diaphragm, a mirror boundary, or a lens rim. The *entrance pupil* is the image of that stop through the optics preceding it, as seen from object space. The *exit pupil* is its image through the optics following it, as seen from image space. Either image can be virtual.

These definitions explain why the pupil diameter is not always the diameter of the first lens. A stop seen through preceding optics can appear magnified. A field stop instead limits which object positions are admitted. *Vignetting* occurs when off-axis bundles are partly blocked; it changes pupil shape and weights as a function of field.

The *chief ray* from a field point passes through the centre of the aperture stop. A *marginal ray* passes through its edge; in a centred system an axial marginal ray describes the aperture cone. Field position describes which object point is being imaged. Pupil position describes which ray from that object point is used. The two must be varied independently to characterize aberrations across an image.

For an object at infinity in air, the f -number is $N = f_{\text{eff}}/D_{\text{entrance}}$. Image-space numerical aperture is $\text{NA} = n' \sin \alpha$, where α is the marginal cone half-angle at the image. In a simple paraxial air system with the pupil appropriately located, $\text{NA} \simeq 1/(2N)$. Finite conjugates, pupil magnification, and high aperture require the actual cone geometry rather than this shortcut.

4.2 Define the ideal before subtracting it

Let an output plane be $z = z_e$, with intended focus $F = (X_0, Y_0, z_f)$ and $L = z_f - z_e > 0$ in a homogeneous output index n' . The ideal converging eikonal on that plane can be written

$$S_{\text{ideal}}(x, y) = C - n' \sqrt{(x - X_0)^2 + (y - Y_0)^2 + L^2}. \quad (4.1)$$

The minus sign is physically necessary: the ideal transverse derivative directs a ray toward the focus. Define the optical path difference by

$$\boxed{W(x, y) = S_{\text{actual}}(x, y) - S_{\text{ideal}}(x, y)}. \quad (4.2)$$

A positive W is a positive excess of eikonal, in length units, at the same evaluation point. The choice of C changes only a constant term called piston. We usually choose it so a chief-ray reference or the pupil mean is zero.

An equivalent computational construction is

$$W = S_{\text{actual}}(P) + n'|F - P| - C.$$

The last straight distance is the distance from the evaluation point to the *chosen ideal focus*. An aberrated physical ray need not travel along that line. The expression subtracts an ideal phase; it must not be described as proving that every physical ray reaches F .

Evaluation on a mirror surface, on a plane, or on a reference sphere gives different coordinate parameterizations. At leading order they often agree well; at higher order the mapping matters. A mode fitted as a function of mirror footprint radius is not automatically the same mode fitted over a planar exit pupil. Record the coordinate surface along with the coefficient table.

4.3 Piston, tilt, and defocus

Adding a constant C_0 to W changes the common phase but not relative phase across a single pupil. This is *piston*. It is irrelevant to the intensity of an isolated coherent image, although differential piston between interferometer arms or mirror segments can matter.

A linear wavefront $W = b_x x + b_y y$ is *tilt*. Its gradient is constant, so it produces a uniform angular change in the paraxial approximation. Tilt shifts an image. Whether it should be removed depends on the question: centring a spot and predicting absolute pointing or distortion are different measurements.

A quadratic radial term is *defocus*. Its connection to changing the reference image plane follows directly from Eq. (4.1). For an on-axis focus,

$$S_{\text{ideal}}(r; L) = \text{piston} - \frac{n'r^2}{2L} + \frac{n'r^4}{8L^3} - \dots$$

Move the reference focus downstream by ΔL , hold the actual eikonal fixed, and expand $1/(L + \Delta L) \simeq 1/L - \Delta L/L^2$. The new difference is

$$W_{\text{new}}(r) = W_{\text{old}}(r) - \frac{n'\Delta L}{2L^2} r^2 + \text{piston} + \text{higher-order terms.} \quad (4.3)$$

Changing the reference focus is therefore not the same operation as adding a positive quadratic phase to the physical wavefront. One changes the subtracted ideal; the other changes the actual optical system.

For normalized radius $\rho = r/a$, a quadratic coefficient B in $W = B\rho^2$ changes under refocusing by $\Delta B = -n'a^2\Delta L/(2L^2)$. The units are consistent: $a^2\Delta L/L^2$ is a length.

4.4 Wavefront slope to ray error: derivation and limits

On a fixed output plane, Eq. (2.6) gives the exact relation between transverse direction cosines and phase:

$$\mathbf{q} = (d_x, d_y) = \frac{1}{n'} \nabla_{\perp} S.$$

Hence $\delta\mathbf{q} = \nabla_{\perp} W/n'$. If both the ideal and actual directions are paraxial, $d_z \simeq 1$, so transverse slope is approximately $\mathbf{q}$. Propagating through axial distance L yields

$$\boxed{\Delta\mathbf{x} \simeq \frac{L}{n'} \nabla_{\perp} W.} \quad (4.4)$$

The sign is positive under Eq. (4.2). A positive x derivative adds positive transverse direction and moves the intercept toward positive x . References defining ideal minus actual OPD have the opposite sign.

For a normalized wavefront $W(u, v)$ with $u = x/a$, $v = y/a$, the correct formula is $\Delta \mathbf{x} \simeq L \nabla_{u,v} W / (n'a)$. As an example, let $W = 20 \text{ nm}(x/a)$, $a = 5 \text{ mm}$, $L = 100 \text{ mm}$, $n' = 1$. Convert $20 \text{ nm} = 2 \times 10^{-5} \text{ mm}$. Then the angle error is 4×10^{-6} radians and the image shift is $0.0004 \text{ mm} = 0.4 \text{ }\mu\text{m}$.

For higher aperture use the actual slope function on the forward branch $d_z > 0$, with $|\mathbf{q}| < 1$:

$$\mathbf{s}(\mathbf{q}) = \frac{\mathbf{q}}{\sqrt{1 - |\mathbf{q}|^2}}.$$

Let $\mathbf{q}_0$ be the ideal transverse direction-cosine vector at the same pupil point. Between fixed planes, exact geometrical propagation gives

$$\Delta \mathbf{x} = L \left[\mathbf{s} \left(\mathbf{q}_0 + \frac{\nabla_{\perp} W}{n'} \right) - \mathbf{s}(\mathbf{q}_0) \right]. \quad (4.5)$$

To first order in a small wavefront perturbation, but without setting the ideal aperture angle to zero, differentiate the slope:

$$\delta \mathbf{s} = \left[\frac{I}{\sqrt{1 - q_0^2}} + \frac{\mathbf{q}_0 \mathbf{q}_0^T}{(1 - q_0^2)^{3/2}} \right] \frac{\nabla_{\perp} W}{n'}.$$

Here $q_0^2 = |\mathbf{q}_0|^2$ and I is the two-dimensional identity matrix. This matrix approaches I paraxially. Higher-order ray coefficients can receive contributions from its aperture dependence, not just from differentiating the next wavefront term. Pupil coordinate transformations and a curved evaluation surface introduce further changes. These distinctions become essential when claiming fifth-order or higher physical ray accuracy.

4.5 Surface displacement is not wavefront error

Move a local reflecting point by displacement $\delta \mathbf{r}$ while keeping nearby incident and outgoing reference endpoints fixed. Differentiating the lengths of the incident and outgoing segments gives

$$\delta \mathcal{L} = n(\mathbf{d}_{\text{in}} - \mathbf{d}_{\text{out}}) \cdot \delta \mathbf{r}.$$

Let the unit normal $\mathbf{N}$ point toward the incident medium and let positive surface height h mean displacement $h\mathbf{N}$. Since $\mathbf{d}_{\text{in}} \cdot \mathbf{N} = -\cos i$ and $\mathbf{d}_{\text{out}} \cdot \mathbf{N} = +\cos i$,

$$\boxed{\delta W = -2nh \cos i.} \quad (4.6)$$

At normal incidence in air the magnitude is twice the surface displacement. The sign says that moving the mirror toward the incoming beam shortens the two-way geometrical path. The relation is a first-order local perturbation with a specified normal displacement and pupil mapping. At oblique incidence a global axial sag is not the same as a normal displacement, and the footprint is distorted.

For a thin transmitted layer between fixed reference planes, replacing external material of index n_e by glass of index n_g through additional near-normal thickness δt gives

$$\boxed{\delta W = (n_g - n_e)\delta t.} \quad (4.7)$$

Thus 10 nm of local mirror height and 10 nm of extra glass thickness do not imply the same phase error. The surface-to-wavefront conversion also depends on the number of passes, incidence angle, refractive index, and spatial mapping. A manufacturer's "surface RMS" cannot be inserted into a wavefront formula without this information.

Chapter 5

Build an exact geometrical calculation you can inspect

5.1 What must be specified

An optical prescription needs surface shapes and vertex positions, material index in each space, clear apertures, input field and wavelength, stop definition, and the evaluation plane or reference sphere. A conic constant by itself is not a mirror prescription. A list of Zernike coefficients without a pupil radius and convention is not a complete wavefront specification.

Use a single consistent length unit internally. Store the physical vacuum wavelength in the same unit when converting OPL to phase. Separate source rays, geometry, index data, and analysis choices so you can change one without silently changing the others. The introductory geometrical sequence is also developed in the university course collection listed as Ref. [1]; the derivations here are presented explicitly for independent study.

5.2 Intersect a ray with a sphere

For sphere centre $\mathbf{c}$ and radius magnitude $|R|$, substitute the ray $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{d}$ into $|\mathbf{r} - \mathbf{c}|^2 = R^2$. Since $|\mathbf{d}| = 1$, expansion gives

$$t^2 + 2bt + c_0 = 0, \quad b = \mathbf{d} \cdot (\mathbf{r}_0 - \mathbf{c}), \quad c_0 = |\mathbf{r}_0 - \mathbf{c}|^2 - R^2.$$

The solutions are $t = -b \pm \sqrt{b^2 - c_0}$. A negative discriminant means the line misses the sphere. But a positive discriminant does not tell you which solution belongs to the intended optical surface. A lens uses one cap of a sphere, not its entire boundary. Select a forward intersection consistent with the cap containing the specified vertex, then check the clear aperture. Close to a surface, exclude a zero-distance self-intersection within a stated numerical tolerance.

For numerical stability, if one root suffers subtraction of nearly equal numbers, compute the more stable root $q = -b - \text{sign}(b)\sqrt{b^2 - c_0}$ and the other as c_0/q , with suitable handling of zero b or q . This follows because the product of the roots is c_0 . Precision lost in a geometric intersection can corrupt a much smaller wavefront difference downstream.

5.3 Intersect a ray with an asphere

For an explicit surface $z = z_v + s(x, y)$, define

$$F(t) = z_0 + td_z - z_v - s(x_0 + td_x, y_0 + td_y).$$

The intersection satisfies $F(t) = 0$. Its derivative is

$$F'(t) = d_z - s_x d_x - s_y d_y.$$

Newton's iteration is $t_{k+1} = t_k - F(t_k)/F'(t_k)$. It comes from replacing F near t_k by its tangent line and finding that line's zero. Use a bracketed or safeguarded method if the derivative becomes small or several intersections are possible. A converged root must still be checked against the intended surface branch and aperture.

At the resulting point, an unnormalized surface normal is $(-s_x, -s_y, 1)$. Normalize it by its Euclidean length and orient it according to the reflection or refraction convention. Surface position and surface slope enter different stages of the ray trace, which is why a small but rapidly varying height error can scatter rays strongly.

5.4 Accumulate OPL and compare on a common plane

For every straight segment of length ℓ_j , add $n_j \ell_j$ before changing the material index at its end. For a collimated input, rays should start on a common plane of equal initial phase, or carry the correct initial eikonal for their different start positions. For a point object, the path from that object to the first intersection supplies the initial optical path.

After the last surface, propagate each physical ray to the same output plane $z = z_e$ using

$$t_e = \frac{z_e - z_{\text{last}}}{d_z}, \quad P_e = P_{\text{last}} + t_e \mathbf{d}.$$

For a real downstream plane t_e should be nonnegative. A virtual pupil may be represented by signed backward continuation in a homogeneous image space; this is a mathematical evaluation, not another physical pass through earlier surfaces.

Let $\mathcal{L}_j(P_e)$ be the accumulated actual eikonal at that point, including the final continuation with the appropriate sign. Choose reference focus F and reference ray c . Then compute

$$W_j = \mathcal{L}_j(P_{e,j}) + n'|F - P_{e,j}| - [\mathcal{L}_c(P_{e,c}) + n'|F - P_{e,c}|]. \quad (5.1)$$

The ray's pupil coordinates should be the coordinates used for the stated fit. If you sample uniformly on an entrance disk and fit on an exit disk, the mapped points need not be uniform in exit area. Use area weights or explicitly label the result as an entrance-pupil parameterization.

5.5 A trace is not verified merely because it produces a spot

Several independent checks test different failure modes:

1. Confirm $|\mathbf{d}| = 1$ after every reflection or refraction and verify the tangential Snell relation at every transmitting surface.
2. Let an entrance ray height tend toward zero. Its focal distance should approach the paraxial matrix result.
3. For a centred rotationally symmetric on-axis system, rotate the input pupil ray and confirm that the output rotates correspondingly.
4. For an ideal paraboloid and axial plane-wave illumination, verify constant phase-to-focus over the admitted aperture.
5. Increase pupil sampling and fitted polynomial order separately. A low residual at sparse sample points is not evidence of an accurate continuous fit.

6. Verify that a fitted coefficient has the predicted aperture scaling in the small-aperture limit. A leading quartic wavefront scales as a^4 before normalization changes.

An optical model can pass one check and fail another. For example, a wrong OPL accumulator can coexist with a correct ray spot because directions were traced correctly. Conversely, an apparently smooth OPD map can be assigned to the wrong pupil coordinates.

5.6 Numerical differentiation and cancellation

The centred finite difference $[W(x + h) - W(x - h)]/(2h)$ approximates $W'(x)$ with leading truncation error proportional to h^2 for a smooth function. Making h arbitrarily small eventually amplifies floating-point noise. Check a sequence of step sizes for a stable region instead of assuming the smallest step is best.

Likewise, subtracting two OPLs of tens of millimetres to recover nanometres involves cancellation. Double precision is often adequate for the examples in these books, but it is not a universal guarantee at picometre accuracy or extreme geometric scales. Use reference-subtracted analytic expressions, compensated accumulation, or higher precision when convergence checks demand it. Test the optical quantity of interest, not just the number of printed digits.

Chapter 6

Foundation exercises and worked solutions

6.1 Exercises

Attempt these before reading the solutions. State units and conventions in every answer.

1. A wavefront has $W = 45$ nm at one pupil point relative to another. Compute the phase difference at 633 nm and 13.5 nm.
2. A spherical surface has $R = 200$ mm and clear radius $a = 10$ mm. Find the quadratic sag at the edge and the first omitted quartic sag. Why can the latter matter even when the quadratic approximation looks accurate on a drawing?
3. Find the mean, RMS after removing the mean, and peak-to-valley value of $W = B\rho^4$ on a uniformly weighted unit disk.
4. A ray enters index 1.5 from air at 30° to the normal. Find its refracted angle. Explain why substituting $\tan i$ for $\sin i$ is not an exact refraction law.
5. Derive the focal length of a thin lens with $n_g = 1.6$, $R_1 = +80$ mm, $R_2 = -120$ mm. Where is the image of an object 300 mm in front of it?
6. A normal-incidence mirror has positive normal height error $h = +8$ nm toward the incoming beam. What is its first-order OPD in air? Compare with a transmitted glass thickness increase of 8 nm at $n_g = 1.5$.
7. At an output plane $a = 10$ mm from its pupil centre to its edge, a reference focus lies $L = 200$ mm downstream in air. How much does the quadratic normalized-pupil coefficient change if the reference focus moves downstream by 0.1 mm?
8. For the thick lens of Chapter 3, use its matrix to find the output ray vector for collimated entrance height 1 mm, then recover its paraxial BFL from the vector.
9. For $W = A\rho^4 + B\rho^2$, choose B to minimize mean-subtracted wavefront RMS. Is the resulting coefficient also guaranteed to minimize every geometrical spot measure?

6.2 Solutions with intermediate steps

1. Phase. Use $\Delta\phi = 2\pi W/\lambda_0$. At 633 nm, $W/\lambda_0 = 45/633 = 0.071090$ and $\Delta\phi = 0.44667$ radians. At 13.5 nm the ratio is $10/3$, so the accumulated difference is $20\pi/3 = 20.94395$ radians.

An interference measurement of wrapped phase only determines this angle modulo 2π ; reconstructing a continuous multiwave OPD requires phase unwrapping or additional information.

2. Sag. The quadratic term is $a^2/(2R) = 100/400 = 0.25$ mm. The quartic term is $a^4/(8R^3) = 10^4/(8 \times 8 \times 10^6) = 0.00015625$ mm = 156.25 nm. It is only 0.0625% of the quadratic sag, yet its normal-incidence reflected path effect has magnitude approximately 312.5 nm under the local perturbation model. Optical phase sensitivity and visual geometric similarity are different criteria.

3. Quartic statistics. The mean is $B\langle\rho^4\rangle = B/3$. The second moment is $B^2\langle\rho^8\rangle = B^2/5$. Therefore the variance is $B^2(1/5 - 1/9) = 4B^2/45$ and $\sigma_W = 2|B|/\sqrt{45}$. The extrema are 0 and B , so peak-to-valley is $|B|$. Removing the mean leaves peak-to-valley unchanged.

4. Refraction. Snell's law gives $\sin t = \sin 30^\circ/1.5 = 1/3$, so $t = \arcsin(1/3) = 19.4712^\circ$. The sine follows from the tangential projection of a *unit* direction and from the path derivative. The tangent is transverse slope per axial distance, a different geometric quantity. Sine and tangent agree to first order near zero, which explains their interchange only in a paraxial approximation.

5. Lens. The power is $0.6(1/80 + 1/120) = 0.0125$ mm⁻¹, so $f = 80$ mm. The image satisfies $1/s' = 1/80 - 1/300 = 11/1200$ mm⁻¹. Thus $s' = 1200/11 = 109.0909$ mm and the magnification is $-s'/s = -4/11$.

6. Surface versus thickness. With the stated mirror height sign, $\delta W = -2h = -16$ nm. The transmitted thickness produces $\delta W = (1.5 - 1)(8 \text{ nm}) = +4$ nm. The signs differ because moving the reflector forward shortens the path, whereas replacing air by more glass increases optical path between fixed planes.

7. Reference focus. Equation (4.3) gives

$$\Delta B = -\frac{(10 \text{ mm})^2(0.1 \text{ mm})}{2(200 \text{ mm})^2} = -0.000125 \text{ mm} = -125 \text{ nm}.$$

The calculation assumes $|\Delta L| \ll L$ and a paraxial pupil. A physical change in lens power requires a separate calculation of how S_{actual} changes.

8. Thick-lens ray. Multiplication gives $y_{\text{out}} = 29/30$ mm and $p_{\text{out}} = -59/3000$. In output air $u \simeq p$, so the distance needed to remove the remaining height is $-y/u = (29/30)(3000/59) = 2900/59$ mm. This agrees with Eq. (3.9). Its dimensions come from the height; reduced angle is dimensionless.

9. Best RMS focus. The variance is a quadratic function of B . Its derivative vanishes when

$$0 = \text{Cov}(A\rho^4 + B\rho^2, \rho^2) = A\left(\frac{1}{4} - \frac{1}{3}\frac{1}{2}\right) + B\left(\frac{1}{3} - \frac{1}{4}\right) = \frac{A+B}{12}.$$

Therefore $B = -A$. Removing the mean yields

$$W - \langle W \rangle = A(\rho^4 - \rho^2 + 1/6).$$

Squaring and using disk moments gives variance $A^2/180$. Hence $\sigma_W = |A|/\sqrt{180}$. This is best mean-square *wavefront* focus under the given weighting. A geometrical metric uses derivatives or extrema and need not have the same optimum. The later chapters explicitly separate these objectives.

Part II

Zernike coefficients: derivation, interpretation, and research use

Chapter 7

What a Zernike coefficient actually means

7.1 From one number to a function of position

A ruler can measure the height of one point on a mirror. It cannot describe the whole mirror with that one measurement. A surface is a function: given two coordinates, the function returns a height. Similarly, a wavefront error is a function that returns an optical path difference at each pupil position. The purpose of a Zernike expansion is to describe such a function using a sequence of numbers, each multiplying a precisely defined spatial pattern.

The phrase “Zernike constants” usually means *Zernike coefficients*. They are constant with respect to position within one fitted map. They need not remain constant when wavelength, field angle, temperature, focus, illumination, aperture, alignment, or the reference surface changes. A mirror does not possess one universal list of Zernike coefficients independent of these choices.

Consider a circular pupil of physical radius a , centred at (x_c, y_c) . Define dimensionless coordinates

$$X = \frac{x - x_c}{a}, \quad Y = \frac{y - y_c}{a}, \quad \rho = \sqrt{X^2 + Y^2}, \quad \theta = \text{atan2}(Y, X). \quad (7.1)$$

The pupil is the disk $0 \leq \rho \leq 1$. Angle increases counterclockwise from positive X toward positive Y when this coordinate plane is viewed in the orientation used to define the map. Looking at a mirror from its opposite side can reverse an axis. Therefore “clockwise on the mirror” is not sufficient metadata; label the actual coordinate axes.

Suppose, for example, that

$$W(X, Y) = 20 \text{ nm} + 30 \text{ nm } X + 40 \text{ nm } (X^2 - Y^2). \quad (7.2)$$

The numbers multiply functions, not points. At the centre the last two terms vanish. At $(X, Y) = (1, 0)$ the value is 90 nm; at $(0, 1)$ it is -20 nm. A coefficient does not usually equal the maximum error or the error at the pupil edge. Its meaning depends on the scale chosen for its multiplying function.

7.2 The distinction between a surface and a wavefront

We will use three different quantities:

$$z(x, y) : \quad \text{geometrical surface sag,} \quad (7.3)$$

$$h(x, y) : \quad \text{surface departure from a specified nominal surface,} \quad (7.4)$$

$$W(x, y) : \quad \text{actual optical eikonal minus ideal eikonal.} \quad (7.5)$$

All three may have length units, but they represent different objects. An eikonal is optical path length accumulated with a specified phase reference. At a common output plane, or on an explicitly chosen common comparison surface,

$$W = S_{\text{actual}} - S_{\text{ideal}}, \quad \phi = \frac{2\pi}{\lambda_0} W, \quad (7.6)$$

where λ_0 is the vacuum wavelength and ϕ is the corresponding phase difference in radians. Optical path includes refractive index: $S = \int n ds$. A positive W means that the actual phase, in the $\exp(ik_0 S - i\omega t)$ convention, has accumulated more than the ideal phase at the same comparison location.

A sag expansion is a geometrical prescription. A wavefront expansion is an optical performance description relative to a chosen ideal wave. Converting one to the other requires the incident field, reflection or refraction law, pupil mapping, and reference wave. Near normal incidence a small mirror height error produces twice the optical path error in magnitude. A lens thickness error instead introduces an index difference. We derive both statements later, including their signs and limitations.

7.3 A basis is a set of building blocks

In a plane, any vector can be written as $v_x \mathbf{e}_x + v_y \mathbf{e}_y$. The unit vectors are building blocks, and v_x, v_y are coordinates. For functions, the building blocks are functions themselves:

$$W(\rho, \theta) = \sum_{n,m} c_n^m Z_n^m(\rho, \theta). \quad (7.7)$$

If the Z_n^m are dimensionless, the coefficients c_n^m have the units of W . A coefficient of $0.05 \mu\text{m}$ is 50 nm ; it is 0.10 waves only when $\lambda_0 = 500 \text{ nm}$. The superscript m is an index, not an exponent of the coefficient.

A finite polynomial expansion represents a finite-dimensional model of a map. It does not assert that every manufactured error is polynomial. Tool marks, scratches, discontinuities at segment boundaries, and unresolved roughness can require many modes or a different representation. A model is useful when its retained coefficients and its residual together answer the design question.

7.4 Averages over a disk and the area element

A small polar patch has radial width $d\rho$ and tangential width $\rho d\theta$, so its area is $\rho d\rho d\theta$. Define

$$\langle f, g \rangle_D = \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(\rho, \theta) g(\rho, \theta) \rho d\rho d\theta. \quad (7.8)$$

The factor $1/\pi$ divides by the disk area. Thus $\langle 1, 1 \rangle_D = 1$. For radial functions,

$$\langle \rho^p, 1 \rangle_D = 2 \int_0^1 \rho^{p+1} d\rho = \frac{2}{p+2}, \quad p > -2. \quad (7.9)$$

For example, the mean of ρ^2 is $1/2$, not $1/3$. The value $1/3$ comes from averaging uniformly in radius, which overweights the centre relative to physical area. This distinction is central to numerical sampling and least-squares fitting.

The mean is $\overline{W} = \langle W, 1 \rangle_D$. The standard deviation of the wavefront, usually called its piston-removed RMS error, is

$$\sigma_W = \sqrt{\langle W - \overline{W}, W - \overline{W} \rangle_D}. \quad (7.10)$$

An unqualified ‘‘RMS’’ is ambiguous: $\sqrt{\langle W, W \rangle_D}$ includes piston, whereas Eq. (7.10) does not. Piston changes the overall phase of a coherent beam but does not change its isolated intensity image. Differential piston between subapertures does change interference and cannot be discarded independently without changing the optical system.

7.5 Why orthogonality makes coefficients useful

Two functions are orthogonal when their inner product is zero. They are orthonormal when, additionally, each has unit squared norm. We will use

$$\langle Z_n^m, Z_{n'}^{m'} \rangle_D = \delta_{nn'} \delta_{mm'}, \quad (7.11)$$

where a Kronecker delta is one for equal indices and zero otherwise. Multiply Eq. (7.7) by $Z_n^{m'}$ and integrate. Every term vanishes except the matching one, giving

$$\boxed{c_n^m = \langle W, Z_n^m \rangle_D}. \quad (7.12)$$

This is a projection formula: the coefficient measures the component of the wavefront parallel to a particular function.

To see why projection is also the best least-squares approximation, let $\widehat{W} = \sum_{j=1}^M c_j Z_j$ and minimize $E = \langle W - \widehat{W}, W - \widehat{W} \rangle_D$. Differentiation gives

$$\frac{\partial E}{\partial c_k} = -2 \langle W, Z_k \rangle_D + 2 \sum_j c_j \langle Z_j, Z_k \rangle_D. \quad (7.13)$$

For an orthonormal basis the sum is simply $2c_k$. Setting the derivative to zero proves Eq. (7.12). The residual is orthogonal to every retained mode. Therefore the error cannot be reduced by adding another amount of a mode that has already been fitted.

Since the piston mode is $Z_0^0 = 1$, an expansion over the complete disk has

$$\overline{W} = c_0^0, \quad \sigma_W^2 = \sum_{(n,m) \neq (0,0)} (c_n^m)^2. \quad (7.14)$$

For a truncated fit the squared norm of the remaining residual must also be included to recover the total error. Equation (7.14) is not generally valid on an annulus, an irregular mask, or a differently weighted pupil when ordinary disk Zernikes are retained.

Chapter 8

Constructing Zernike polynomials from scratch

8.1 Why the angular functions are sines and cosines

A point returns to itself when θ increases by 2π ; a physical map must have the same value there. The functions $\cos(m\theta)$ and $\sin(m\theta)$ satisfy this condition for integer m . They also have particularly simple rotation laws and orthogonality:

$$\int_0^{2\pi} \cos(m\theta) \cos(m'\theta) d\theta = \pi \delta_{mm'} \quad (m, m' > 0), \quad (8.1)$$

$$\int_0^{2\pi} \sin(m\theta) \sin(m'\theta) d\theta = \pi \delta_{mm'} \quad (m, m' > 0), \quad (8.2)$$

$$\int_0^{2\pi} \sin(m\theta) \cos(m'\theta) d\theta = 0. \quad (8.3)$$

The constant angular function has squared integral 2π instead of π . This factor of two will explain the different normalization of rotationally symmetric and non-symmetric modes.

The number $|m|$ counts angular cycles around the pupil. It does not count radial rings and does not by itself specify the degree of a polynomial. For example, $\rho^2 \cos 2\theta = X^2 - Y^2$ has two angular cycles and degree two. The identity

$$(X + iY)^\ell = \rho^\ell (\cos \ell\theta + i \sin \ell\theta) \quad (8.4)$$

shows why a regular Cartesian polynomial with angular frequency ℓ begins with ρ^ℓ . Multiplication by $\rho^2 = X^2 + Y^2$ preserves that angular frequency. Thus the allowed radial powers are $\rho^\ell, \rho^{\ell+2}, \rho^{\ell+4}, \dots$

8.2 Gram–Schmidt: remove what has already been represented

We first construct rotationally symmetric modes. Start with 1 and then ρ^2 . Subtract the mean of the second:

$$q_2 = \rho^2 - \langle \rho^2, 1 \rangle_D = \rho^2 - \frac{1}{2}. \quad (8.5)$$

Its norm is

$$\langle q_2, q_2 \rangle_D = \langle \rho^4, 1 \rangle_D - \langle \rho^2, 1 \rangle_D + \frac{1}{4} \quad (8.6)$$

$$= \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12}. \quad (8.7)$$

Division by $1/\sqrt{12}$ therefore gives

$$Z_2^0 = \sqrt{3}(2\rho^2 - 1). \quad (8.8)$$

This operation is Gram–Schmidt orthogonalization. Its physical meaning here is that the new pattern contains no mean optical path.

Now start with ρ^4 and remove its projections onto 1 and Z_2^0 :

$$q_4 = \rho^4 - \frac{1}{3} - \langle \rho^4, Z_2^0 \rangle_D Z_2^0, \quad (8.9)$$

$$\langle \rho^4, Z_2^0 \rangle_D = \sqrt{3} \left(2\frac{1}{4} - \frac{1}{3} \right) = \frac{\sqrt{3}}{6}, \quad (8.10)$$

$$q_4 = \rho^4 - \rho^2 + \frac{1}{6}. \quad (8.11)$$

Squaring this polynomial and using Eq. (7.9) gives $\langle q_4, q_4 \rangle_D = 1/180$. Consequently,

$$Z_4^0 = \sqrt{5}(6\rho^4 - 6\rho^2 + 1). \quad (8.12)$$

The lower powers are essential. A pure quartic term is not orthogonal to piston or focus. The Zernike mode commonly called primary spherical aberration already contains the lower-order terms that make it orthogonal to both.

For a nonzero angular frequency, consider $\rho \cos \theta = X$. Its mean square is $1/4$, so $Z_1^1 = 2\rho \cos \theta = 2X$. To construct a higher mode with the same angular frequency, begin with $\rho^3 \cos \theta$ and remove the tilt projection:

$$\langle \rho^3 \cos \theta, 2\rho \cos \theta \rangle_D = 2 \int_0^1 \rho^5 d\rho = \frac{1}{3}, \quad (8.13)$$

$$q_3 = \left(\rho^3 - \frac{2}{3}\rho \right) \cos \theta. \quad (8.14)$$

Its squared norm is $1/72$. The normalized result is

$$Z_3^1 = \sqrt{8}(3\rho^3 - 2\rho) \cos \theta. \quad (8.15)$$

This is balanced coma. The subtraction of a tilt term is mathematical orthogonalization; it does not mean that a physical coma-producing surface necessarily introduces an equal and opposite mechanical tilt.

8.3 A general construction using one new variable

Fix the nonnegative angular frequency $\ell = |m|$ and write $n = \ell + 2k$ with $k = 0, 1, 2, \dots$. Introduce $u = \rho^2$ and seek

$$R_n^\ell(\rho) = \rho^\ell Q_k(u). \quad (8.16)$$

The radial inner product becomes

$$\int_0^1 R_n^\ell R_{n'}^\ell \rho d\rho = \frac{1}{2} \int_0^1 u^\ell Q_k(u) Q_{k'}(u) du. \quad (8.17)$$

We therefore need ordinary polynomials orthogonal on $[0, 1]$ with weight u^ℓ . A convenient construction is

$$Q_k(u) = \frac{1}{k! u^\ell} \frac{d^k}{du^k} \left[u^{k+\ell} (u-1)^k \right]. \quad (8.18)$$

The apparent division by u^ℓ does not produce a singular function: each differentiated term still contains that power. This is a shifted Jacobi Rodrigues formula; the general Jacobi conventions are documented by NIST [3].

We can prove orthogonality without assuming any theory of special functions. Let $p_j(u)$ be any polynomial of degree $j < k$. Substitute Eq. (8.18):

$$\int_0^1 u^\ell Q_k p_j du = \frac{1}{k!} \int_0^1 p_j \frac{d^k}{du^k} [u^{k+\ell}(u-1)^k] du. \quad (8.19)$$

Integrate by parts once. The derivative transfers to p_j , with a minus sign. The boundary term vanishes because the bracket has a zero of order at least k at both endpoints. Repeat k times. All boundary terms vanish for the same reason, leaving

$$\frac{(-1)^k}{k!} \int_0^1 \frac{d^k p_j}{du^k} u^{k+\ell}(u-1)^k du = 0. \quad (8.20)$$

The last equality follows because the k th derivative of a polynomial of degree less than k is zero. Hence Q_k is orthogonal to every earlier polynomial, including $Q_0, \dots, Q_{k-1}$.

8.4 Deriving the normalization instead of memorizing it

The leading coefficient of Q_k is

$$C_k = \frac{(2k+\ell)!}{k!(k+\ell)!}. \quad (8.21)$$

Indeed, the highest term inside the derivative in Eq. (8.18) is $u^{2k+\ell}$. After k derivatives and division by $k!u^\ell$, the displayed coefficient remains. Using $p = Q_k$ in the integration-by-parts calculation now gives

$$\int_0^1 u^\ell Q_k^2 du = \frac{(-1)^k}{k!} \int_0^1 Q_k^{(k)} u^{k+\ell}(u-1)^k du \quad (8.22)$$

$$= C_k \int_0^1 u^{k+\ell}(1-u)^k du \quad (8.23)$$

$$= C_k \frac{(k+\ell)!k!}{(2k+\ell+1)!} = \frac{1}{2k+\ell+1} = \frac{1}{n+1}. \quad (8.24)$$

The factorial integral in the third line can itself be proved by repeated integration by parts, starting with $\int_0^1 u^p du = 1/(p+1)$. Returning to ρ gives

$$\int_0^1 [R_n^\ell(\rho)]^2 \rho d\rho = \frac{1}{2(n+1)}. \quad (8.25)$$

Combining this result with the angular integrals leads to our complete definition:

$$Z_n^m(\rho, \theta) = \begin{cases} \sqrt{n+1} R_n^0(\rho), & m = 0, \\ \sqrt{2(n+1)} R_n^m(\rho) \cos(m\theta), & m > 0, \\ \sqrt{2(n+1)} R_n^{|m|}(\rho) \sin(|m|\theta), & m < 0. \end{cases} \quad (8.26)$$

The allowed pairs obey $n \geq 0$, $|m| \leq n$, and $n - |m|$ even. Negative m is our label for a sine, not an instruction to evaluate $\sin(m\theta)$ with a negative argument. This distinction prevents an otherwise easy sign error.

8.5 The factorial formula and Jacobi connection

Expand $(u-1)^k$ by the binomial theorem in Eq. (8.18), differentiate each power, and set $u = \rho^2$. Relabelling the finite sum gives

$$R_n^\ell(\rho) = \sum_{s=0}^{(n-\ell)/2} \frac{(-1)^s (n-s)!}{s! \left(\frac{n+\ell}{2} - s\right)! \left(\frac{n-\ell}{2} - s\right)!} \rho^{n-2s}. \quad (8.27)$$

This formula is explicit and useful at low order. At high order the individual terms can be enormous while their sum is small, so floating-point cancellation becomes serious. Recurrences or stable Jacobi evaluators are preferable for high radial degree.

In the standard Jacobi convention,

$$R_{\ell+2k}^\ell(\rho) = \rho^\ell P_k^{(0,\ell)}(2\rho^2 - 1). \quad (8.28)$$

The substitution $x = 2u - 1$ converts the Jacobi weight $(1-x)^0(1+x)^\ell$ into a constant multiple of u^ℓ . It also gives $R_n^\ell(1) = 1$. Thus a library routine must be called with parameters $(0, \ell)$ and argument $2\rho^2 - 1$ for this particular identity. Swapping the parameters without also changing the argument and sign gives a different polynomial.

8.6 A complete radial table through degree six

In Table 8.1, each nonzero ℓ generates both a cosine and a sine mode using Eq. (8.26). All modes through $n = 6$ are therefore specified; there are $1 + 2 + 3 + 4 + 5 + 6 + 7 = 28$ real modes. The descriptive names are aids to recognition, whereas (n, m) and the explicit formula define the function.

Table 8.1: Radial polynomials through $n = 6$.

n	ℓ	$R_n^\ell(\rho)$
0	0	1
1	1	ρ
2	0	$2\rho^2 - 1$
2	2	ρ^2
3	1	$3\rho^3 - 2\rho$
3	3	ρ^3
4	0	$6\rho^4 - 6\rho^2 + 1$
4	2	$4\rho^4 - 3\rho^2$
4	4	ρ^4
5	1	$10\rho^5 - 12\rho^3 + 3\rho$
5	3	$5\rho^5 - 4\rho^3$
5	5	ρ^5
6	0	$20\rho^6 - 30\rho^4 + 12\rho^2 - 1$
6	2	$15\rho^6 - 20\rho^4 + 6\rho^2$
6	4	$6\rho^6 - 5\rho^4$
6	6	ρ^6

For example,

$$Z_2^2 = \sqrt{6}(X^2 - Y^2), \quad Z_2^{-2} = 2\sqrt{6}XY, \quad (8.29)$$

$$Z_3^1 = \sqrt{8}[3(X^2 + Y^2) - 2]X, \quad Z_3^{-1} = \sqrt{8}[3(X^2 + Y^2) - 2]Y, \quad (8.30)$$

$$Z_3^3 = \sqrt{8}(X^3 - 3XY^2), \quad Z_3^{-3} = \sqrt{8}(3X^2Y - Y^3). \quad (8.31)$$

Cartesian forms are useful when differentiating at the origin, where polar-coordinate derivatives contain removable singular expressions.

8.7 Single-index systems and the danger of a bare number

Optical software often replaces (n, m) by one index j . Noll, OSA/ANSI, Fringe, and software-specific orderings do not all assign the same functions to the same integer. They can also use different normalizations or signs. Therefore a statement such as “coefficient 8 is 0.2” is incomplete.

To exchange coefficients safely, export a table containing n , m , the polynomial or normalization, units, aperture radius, coordinate orientation, OPD sign, reference wave, and removed terms. To convert from a function $F_j = \alpha_j Z_n^m$ carrying coefficient b_j , use $c_n^m = \alpha_j b_j$ after matching the actual functions. If the other package reverses the sine convention, the relevant α_j is negative. Never infer a conversion solely from a familiar-looking aberration name. The optical-metrology discussion by Wyant and Creath also emphasizes convention and sampling issues [2].

Chapter 9

Reading a coefficient table as an optical diagnosis

9.1 Piston, tilt, and focus are choices as well as errors

For a full disk, c_0^0 is the mean OPD. The next two modes are $2X$ and $2Y$, representing a plane wave tilted relative to the reference. Their physical gradients are

$$\frac{\partial W_{\text{tilt}}}{\partial x} = \frac{2c_1^1}{a}, \quad \frac{\partial W_{\text{tilt}}}{\partial y} = \frac{2c_1^{-1}}{a}. \quad (9.1)$$

At a forward output plane in medium n' , the paraxial ray direction change is the gradient divided by n' . At a distance L this gives

$$\Delta x = \frac{2L}{n'a} c_1^1, \quad \Delta y = \frac{2L}{n'a} c_1^{-1}. \quad (9.2)$$

The sign is positive for the actual-minus-ideal convention: a positive x gradient increases the forward ray's x direction cosine. This follows from $\mathbf{s} = \nabla S/n'$; it should not be copied into a different OPD convention without changing signs.

Defocus changes quadratic phase curvature. At an output plane $z = 0$, an ideal converging spherical wave whose centre is $(0, 0, L)$ has, after choosing zero phase at the pupil centre,

$$S_{\text{id},L}(r) = n' \left[L - \sqrt{L^2 + r^2} \right] = -\frac{n'r^2}{2L} + \frac{n'r^4}{8L^3} + \dots. \quad (9.3)$$

Changing the reference focus to $L + \delta L$ gives

$$S_{\text{id},L+\delta L} - S_{\text{id},L} \simeq \frac{n'r^2}{2L^2} \delta L. \quad (9.4)$$

For fixed actual wavefront, the residual $W = S_{\text{actual}} - S_{\text{id}}$ therefore changes by the *negative* of this quantity. Since $r^2 = a^2[1 + Z_2^0/\sqrt{3}]/2$,

$$\delta c_2^0 = -\frac{n'a^2}{4\sqrt{3}L^2} \delta L, \quad \delta L_{\text{remove}} = \frac{4\sqrt{3}L^2}{n'a^2} c_2^0. \quad (9.5)$$

This is a small-shift, paraxial relation with the pupil held fixed. It explains why removing defocus numerically corresponds approximately to refocusing physically. Moving a detector or element in a real system can also change pupil mapping and higher-order aberrations.

9.2 Astigmatism and the orientation of a paired mode

Let $a_c = c_2^2$ and $a_s = c_2^{-2}$. Then

$$W_{\text{ast}} = \sqrt{6}\rho^2[a_c \cos 2\theta + a_s \sin 2\theta]. \quad (9.6)$$

Define $A = \sqrt{a_c^2 + a_s^2}$ and $\psi = \frac{1}{2} \text{atan2}(a_s, a_c)$. Using the cosine subtraction identity gives

$$W_{\text{ast}} = \sqrt{6}A\rho^2 \cos[2(\theta - \psi)]. \quad (9.7)$$

The orientation is ambiguous by π , as expected for an unoriented principal axis. Exchanging which axis is called positive also changes the sign representation. The quantity A is the RMS contribution of the pair on the full disk and is invariant under rotation.

Near the centre this wavefront has different quadratic curvatures along its two principal axes. Rays in the two corresponding meridional planes therefore focus at different axial distances. This explains the association with astigmatism. At a single field point the coefficient does not identify whether the cause was a tilted element, a cylindrical surface error, system design, or stress.

9.3 Coma, spherical aberration, and balanced polynomials

The lowest coma pair has $n = 3$, $|m| = 1$. It has one angular cycle but an additional radial structure compared with tilt. The primary spherical mode has $n = 4$, $m = 0$. Higher radial members of these families are often called secondary coma, secondary spherical aberration, and so forth. These names identify pupil patterns at a fixed field point.

Classical aberration *order* is a different bookkeeping system. A quartic wavefront term has a cubic gradient and contributes to third-order ray aberration. Moreover, a fourth-degree joint expansion in pupil and field includes coma and astigmatism terms with lower pupil degree. Thus Zernike radial degree n is neither automatically Seidel order nor ray order. The companion aberrations textbook develops the joint pupil–field expansion explicitly.

The following exact algebraic identities translate common radial monomials into our normalized modes:

$$\rho^2 = \frac{1}{2} + \frac{Z_2^0}{2\sqrt{3}}, \quad (9.8)$$

$$\rho^4 = \frac{1}{3} + \frac{Z_2^0}{2\sqrt{3}} + \frac{Z_4^0}{6\sqrt{5}}, \quad (9.9)$$

$$\rho^6 = \frac{1}{4} + \frac{9Z_2^0}{20\sqrt{3}} + \frac{Z_4^0}{4\sqrt{5}} + \frac{Z_6^0}{20\sqrt{7}}. \quad (9.10)$$

For the last identity, solve $R_6^0 = 20\rho^6 - 30\rho^4 + 12\rho^2 - 1$ for ρ^6 , substitute the preceding two identities, and collect coefficients. The resulting constant is $1/4$, as independently required by the disk mean.

Consequently, if

$$W = A\rho^4 + B\rho^6 + D\rho^2 + C, \quad (9.11)$$

then

$$c_0^0 = C + \frac{D}{2} + \frac{A}{3} + \frac{B}{4}, \quad (9.12)$$

$$c_2^0 = \frac{D + A}{2\sqrt{3}} + \frac{9B}{20\sqrt{3}}, \quad (9.13)$$

$$c_4^0 = \frac{A}{6\sqrt{5}} + \frac{B}{4\sqrt{5}}, \quad (9.14)$$

$$c_6^0 = \frac{B}{20\sqrt{7}}. \quad (9.15)$$

These equations are useful in design because a sixth-power sag or phase term also changes lower radial Zernike coefficients. Controlling c_6^0 independently generally requires compensating lower powers.

9.4 A numerical coefficient budget

Suppose the nonzero coefficients after removing piston, tilt, and focus are $c_2^2 = 30$ nm, $c_2^{-2} = 40$ nm, $c_3^1 = 20$ nm, and $c_4^0 = -10$ nm. The astigmatism pair contributes 50 nm RMS, and the full retained map has

$$\sigma_W = \sqrt{30^2 + 40^2 + 20^2 + (-10)^2} \text{ nm} = 54.77 \text{ nm}. \quad (9.16)$$

The sign matters for the shape and for compensation, but not for the individual squared contribution. At $\lambda_0 = 550$ nm the error is 0.0996 waves RMS. If the unmodelled residual independently orthogonal to the fitted modes has RMS 15 nm, the total is $\sqrt{3000 + 225} = 56.79$ nm. Reporting only 54.77 nm would omit measured structure.

Chapter 10

Zernike coefficients of mirrors

10.1 Deriving the reflection factor with reference planes

Consider a small planar mirror patch. Let its unit normal $\mathbf{N}$ point into the incident medium. An incident ray has unit propagation direction $\mathbf{s}_i$ with $\mathbf{s}_i \cdot \mathbf{N} = -\cos i$, where i is measured from the normal. Its reflected direction is

$$\mathbf{s}_r = \mathbf{s}_i - 2(\mathbf{s}_i \cdot \mathbf{N})\mathbf{N}, \quad \mathbf{s}_r \cdot \mathbf{N} = +\cos i. \quad (10.1)$$

Move the reflecting patch by $h\mathbf{N}$, with $h > 0$ toward the incident medium. Keep the incident and outgoing phase-reference planes fixed. The first-order change of the incoming optical path is $n\mathbf{s}_i \cdot (h\mathbf{N})$; the change of the outgoing path is $-n\mathbf{s}_r \cdot (h\mathbf{N})$. The minus sign appears because moving the starting point of the outgoing segment forward shortens that segment. Adding the changes gives

$$\boxed{\delta W = n(\mathbf{s}_i - \mathbf{s}_r) \cdot h\mathbf{N} = -2nh \cos i.} \quad (10.2)$$

At normal incidence the raised mirror makes both legs shorter, so a positive height gives a negative OPD. This is a direct sign check. Adding the lengths of two oblique segments as $-2nh/\cos i$ would compare inconsistent endpoints; their lateral movement must also be accounted for. The reference-plane derivation gives the correct cosine factor for the phase change.

More generally, at a stationary optical path the first-order change due to displacement of an interface point is

$$\delta S = (n_i \mathbf{s}_i - n_o \mathbf{s}_o) \cdot \delta \mathbf{r}. \quad (10.3)$$

The first-order changes of neighbouring ray segments cancel at an unperturbed stationary path; changes of the ray geometry contribute at higher order. This is why a local height sensitivity can be useful even though a displaced surface also changes ray directions.

10.2 When surface and wavefront coefficients are proportional

If the incidence is approximately normal and uniform, the same disk coordinates describe surface and pupil, the deformation is small, and coating phase is unchanged, expand

$$h = \sum b_n^m Z_n^m, \quad \delta W = -2n \sum b_n^m Z_n^m. \quad (10.4)$$

Thus

$$\boxed{\delta c_n^m = -2nb_n^m} \quad \text{at near-normal incidence.} \quad (10.5)$$

Here b_n^m are surface coefficients and c_n^m are OPD coefficients. The minus sign is tied to our positive surface height direction. If the incidence is constant but oblique, use $-2n \cos i$ provided the coordinate mapping has been specified consistently. If incidence varies across the footprint,

$$\delta c_j = -2n \sum_k b_k \langle Z_j, \cos i(\rho, \theta) Z_k \rangle_D. \quad (10.6)$$

Multiplication by a nonconstant sensitivity mixes modes. A single surface mode can then produce several wavefront modes.

Suppose a mirror in air has $b_2^2 = 12 \text{ nm}$, $b_3^1 = -8 \text{ nm}$, and $b_4^0 = 5 \text{ nm}$, with the stated positive normal pointing toward the incident beam. Near normal incidence its OPD coefficients are $-24, +16, -10 \text{ nm}$. The surface RMS is $\sqrt{144 + 64 + 25} = 15.26 \text{ nm}$, and the OPD RMS is 30.53 nm . An interferometer report of “30.53 nm RMS” must state whether it already applied the reflection conversion; applying the factor of two twice is a common metrology error.

10.3 From curvature and conic constant to sag

A rotationally symmetric conic surface with vertex at $z = 0$ can be defined implicitly by

$$r^2 - 2Rz + (1 + K)z^2 = 0, \quad (10.7)$$

where R is the vertex radius and K is the conic constant in this specified coordinate convention. Solve the quadratic for the branch passing through the vertex and rationalize its numerator:

$$z(r) = \frac{r^2}{R \left[1 + \sqrt{1 - (1 + K)r^2/R^2} \right]}. \quad (10.8)$$

This expression is also well behaved for $K = -1$, where it reduces to $z = r^2/(2R)$. Expanding the square root gives

$$z = \frac{r^2}{2R} + \frac{(1 + K)r^4}{8R^3} + \frac{(1 + K)^2 r^6}{16R^5} + O(r^8). \quad (10.9)$$

A sphere has $K = 0$, and a paraboloid has $K = -1$. An additional aspheric prescription may contain $A_4 r^4 + A_6 r^6 + \dots$. If r and z are in millimetres, A_4 has units mm^{-3} and A_6 has units mm^{-5} . They are not normalized Zernike coefficients.

For a surface departure $h = A_4 r^4 + A_6 r^6$, write $A = A_4 a^4$ and $B = A_6 a^6$, then apply Eq. (9.15) to obtain surface coefficients. This shows explicitly why changing the specified aperture changes the numbers even when the physical surface is unchanged.

10.4 Why a paraboloid focuses an axial plane wave exactly

Choose a meridional coordinate system in air. The incident plane wave travels in the $-z$ direction from a plane $z = z_{\text{in}} > 0$. The mirror is a bowl opening toward positive z , with vertex at zero. Let its desired focus be $F = (0, f)$, $f > 0$. At surface point $P = (r, z)$ the incident path from the reference plane is $z_{\text{in}} - z$. The distance from that point to F is

$$d = \sqrt{r^2 + (f - z)^2}. \quad (10.10)$$

Equal total path to the focus requires $z_{\text{in}} - z + d = z_{\text{in}} + f$, or $d = f + z$. Squaring this equality gives

$$r^2 + (f - z)^2 = (f + z)^2 \implies z = \frac{r^2}{4f}. \quad (10.11)$$

The constant-path surface is a paraboloid with $R = 2f$. Differentiating the constant-path condition along the surface makes the tangential component of $\mathbf{s}_i - \mathbf{s}_r$ vanish; that is exactly the reflection law. Thus equal path and the correct reflected ray direction agree. For an on-axis plane wave, ideal geometrical paraboloidal reflection has no residual wavefront error relative to this focus. It can still have finite-aperture diffraction and off-axis aberrations.

10.5 Sphere versus paraboloid at the fixed Gaussian focus

Use the same geometry but replace the mirror by a sphere of radius R :

$$z_s = R - \sqrt{R^2 - r^2}, \quad f = R/2. \quad (10.12)$$

On the *common mirror surface* the actual incident/reflected eikonal is $-z_s$ up to a constant, while the ideal converging eikonal is $f - \sqrt{r^2 + (f - z_s)^2}$. Their difference is

$$W_{\text{surf}}(r) = -z_s + \sqrt{r^2 + (f - z_s)^2} - f. \quad (10.13)$$

The distance to F belongs to the *ideal comparison wave*; this equation does not assume that actual spherical-mirror rays reach F . It is the optical path function sampled at surface-footprint radius r .

The sphere identity $r^2 = 2Rz_s - z_s^2$ gives, at $f = R/2$,

$$\sqrt{r^2 + (f - z_s)^2} = \frac{R}{2} \sqrt{1 + 4z_s/R}. \quad (10.14)$$

Using $\sqrt{1+v} = 1 + v/2 - v^2/8 + v^3/16 + \dots$,

$$W_{\text{surf}} = -\frac{z_s^2}{R} + \frac{2z_s^3}{R^2} + O(z_s^4), \quad (10.15)$$

$$z_s = \frac{r^2}{2R} + \frac{r^4}{8R^3} + \frac{r^6}{16R^5} + O(r^8), \quad (10.16)$$

$$\boxed{W_{\text{surf}}(r) = -\frac{r^4}{4R^3} + \frac{r^6}{8R^5} + O(r^8).} \quad (10.17)$$

The quartic sign also follows from a simple perturbation argument: the sphere lies above the paraboloid by $r^4/(8R^3)$ to leading order; a positive mirror height gives a negative OPD twice that size.

10.6 Why a pupil plane must be named at sixth order

An exact eikonal can be evaluated on different surfaces, but the transverse coordinates do not generally stay the same along a ray. Consequently, a coefficient obtained by fitting surface radius need not equal one obtained by fitting the outgoing ray's intercept on a plane. Even the ideal-wave subtraction is evaluated at a different point. These distinctions enter at the order that a detailed design calculation may be trying to retain.

For the sphere just considered, set $q = \sqrt{R^2 - r^2}$ so $z_s = R - q$. An upward-pointing unit normal is $\mathbf{N} = (-r/R, q/R)$. Reflecting $\mathbf{s}_i = (0, -1)$ gives

$$s_x = -\frac{2rq}{R^2}, \quad s_z = 1 - \frac{2r^2}{R^2}. \quad (10.18)$$

Extrapolate the outgoing ray to the vertex plane $z = 0$. Its transverse coordinate and signed propagation distance are

$$x = r - z_s \frac{s_x}{s_z}, \quad \ell = -\frac{z_s}{s_z}. \quad (10.19)$$

This plane is a *virtual comparison plane*: the propagation is backwards along the reflected ray for a bowl with positive sag. The actual eikonal there is $S_a = -z_s + \ell$ and the ideal eikonal is $S_i = f - \sqrt{f^2 + x^2}$. Therefore

$$W_{z=0}(x) = -z_s - \frac{z_s}{s_z} - f + \sqrt{f^2 + x^2}, \quad (10.20)$$

with r eliminated using the intercept equation. Expanding gives

$$x = r + \frac{r^3}{R^2} + O(r^5), \quad \boxed{W_{z=0}(x) = -\frac{x^4}{4R^3} + \frac{9x^6}{8R^5} + O(x^8)}. \quad (10.21)$$

The leading quartic term agrees with Eq. (10.17); the sixth-order term differs. Neither expression can be used without its sampling definition. For a physical output plane ahead of the mirror, trace to that plane and repeat the common-point eikonal subtraction. For an exit-pupil analysis, use the actual exit-pupil mapping of the system. The formulas in this section also assume rays remain in the chosen forward branch, in particular $s_z > 0$ for this example.

10.7 A worked mirror design estimate and balanced focus

Take a 100 mm diameter spherical mirror in air with $R = 1000$ mm and Gaussian focal length $f = 500$ mm. For a quick paraxial estimate let $a = 50$ mm and retain only the quartic term:

$$W = Ap^4, \quad A = -\frac{a^4}{4R^3} = -0.0015625 \text{ mm} = -1562.5 \text{ nm}. \quad (10.22)$$

The coefficient A is the edge value in this monomial representation; it is not the spherical Zernike coefficient. Equation (9.15) gives

$$c_0^0 = -520.833 \text{ nm}, \quad (10.23)$$

$$c_2^0 = -451.055 \text{ nm}, \quad (10.24)$$

$$c_4^0 = -116.462 \text{ nm}. \quad (10.25)$$

At the fixed Gaussian reference, after piston removal only, $\sigma_W = |A|\sqrt{4/45} = 465.847$ nm. The pupil pattern includes a large focus component because ρ^4 is not balanced.

If axial focus is adjustable, add $D\rho^2$ and minimize RMS after piston removal. Setting $c_2^0 = 0$ gives $D = -A$, leaving

$$W_{\text{balanced}} = A \left(\rho^4 - \rho^2 + \frac{1}{6} \right) = \frac{A}{6\sqrt{5}} Z_4^0. \quad (10.26)$$

Its RMS is 116.462 nm. The physical focus shift estimated from Eq. (9.5) is

$$\delta L = \frac{4\sqrt{3}f^2}{a^2} c_2^0 = -0.3125 \text{ mm}. \quad (10.27)$$

The negative sign means the best paraxial balance lies closer to the mirror than the Gaussian focus. At this aperture, an exact best-focus calculation will have higher-order corrections. “Balanced” here means minimum phase variance in the stated approximation; it need not give the minimum geometrical spot RMS or the best image for a large error.

The sixth-order surface-footprint monomial coefficient is $B = a^6/(8R^5) = 1.953125 \text{ nm}$. Its additions to $(c_0^0, c_2^0, c_4^0, c_6^0)$ are

$$\left(\frac{B}{4}, \frac{9B}{20\sqrt{3}}, \frac{B}{4\sqrt{5}}, \frac{B}{20\sqrt{7}} \right). \quad (10.28)$$

These are approximately $(0.4883, 0.5074, 0.2184, 0.0369) \text{ nm}$. On a vertex-plane disk with the same numerical radius, the sixth-power coefficient would instead be $9B$. This is a comparison of two coordinate specifications, not a ninefold change of the physical mirror.

10.8 Correcting a measured wavefront with a mirror

If a desired OPD correction is $W_{\text{corr}} = -W_{\text{err}}$, Eq. (10.2) gives, near normal incidence in air,

$$h_{\text{command}} = \frac{W_{\text{err}}}{2}. \quad (10.29)$$

A positive measured OPD requires raising the surface toward the incident medium to shorten the path. For a deformable mirror, however, the actuators do not usually create isolated Zernike modes. If actuator k produces calibrated surface influence function g_k , a command vector u_k creates $h = \sum_k u_k g_k$. Fit or optimize these influence functions through the optical sensitivity, subject to stroke and force limits. A Zernike target is a compact description of the desired wavefront; it is not by itself an actuator command.

Chapter 11

Zernike coefficients of lenses

11.1 Deriving the optical path of a thin thickness perturbation

Put a transparent piece of refractive index n_g in an external medium of index n_e . Compare two nearly axial paths between the same input and output planes, separated by distance d . If the glass thickness at transverse position (x, y) is t , the phase-screen approximation assigns the optical path

$$S = n_e(d - t) + n_g t = n_e d + (n_g - n_e)t. \quad (11.1)$$

Increasing thickness by δt , while replacing an equal axial length of external medium, changes the optical path by

$$\boxed{\delta W = (n_g - n_e)\delta t.} \quad (11.2)$$

For ordinary visible-light glass in air the index difference is positive, so extra thickness increases OPD. This sign differs from raising a mirror toward the incident medium.

The derivation assumes a thin phase screen, nearly axial paths, and a common transverse coordinate. For a thick powered lens, the actual ray travels at an angle in glass, intersects the second surface at another height, and changes its external propagation path. The total wavefront of that lens cannot in general be calculated by multiplying its nominal thickness function by $n_g - n_e$. The simple formula is most reliable for small thickness errors on a nearly normal transmitting element or as a deliberately stated thin-screen model.

For an interface displaced normally toward the incident medium, Eq. (10.3) gives the more general first-order sensitivity

$$\delta W = (n_o \cos t - n_i \cos i)h, \quad (11.3)$$

where i and t are measured from the corresponding surface normal. The surface displacement must be normal displacement; axial sag change requires its projection on that normal. Both surfaces of a lens can contribute, with their own signs and ray mappings.

11.2 A lens thickness-error example

Suppose a weak transmitting element in air has $n_g = 1.50$ at the measurement wavelength. Its thickness departure on a circular pupil is

$$\delta t = (40 \text{ nm})Z_2^2 - (20 \text{ nm})Z_4^0. \quad (11.4)$$

The induced OPD coefficients are $c_2^2 = 20 \text{ nm}$ and $c_4^0 = -10 \text{ nm}$, so $\sigma_W = 22.36 \text{ nm}$. If the same numerical map instead described the height of a mirror in the normal-incidence convention, the

coefficients would be -80 and $+40$ nm and the RMS would be 89.44 nm. The same length-valued input map can therefore have very different optical consequences.

For a real lens, thickness departure is a difference of surface maps. If both maps use the same forward axial coordinate, $t = z_2 - z_1$, hence $\delta t = \delta z_2 - \delta z_1$. If a metrology instrument instead reports both surfaces with their outward normals as positive, one normal points opposite the other. Convert those signs before subtracting surface coefficients.

11.3 Deriving the ideal focusing phase screen

Let a plane wave reach a zero-thickness screen at $z = 0$ in output medium n_e . We want its transmitted rays to converge to $F = (0, 0, f)$. The optical path from screen radius r to the focus is $n_e \sqrt{f^2 + r^2}$. To make total phase at F independent of r , the screen must add the relative eikonal

$$T_{\text{ideal}}(r) = n_e \left[f - \sqrt{f^2 + r^2} \right]. \quad (11.5)$$

The rim receives less delay than the centre because its subsequent path to the focus is longer. Expanding yields

$$T_{\text{ideal}} = -\frac{n_e r^2}{2f} + \frac{n_e r^4}{8f^3} - \frac{n_e r^6}{16f^5} + O(r^8). \quad (11.6)$$

Within the straight-through thickness approximation, this phase can be realized by

$$t(r) = t_0 + \frac{n_e}{n_g - n_e} \left[f - \sqrt{f^2 + r^2} \right]. \quad (11.7)$$

The constant t_0 sets a mechanically positive thickness but contributes only piston in the idealized phase model. This is a design for a *phase screen*. It is not a proof that an arbitrary finite lens with this thickness and unspecified surface shapes has perfect imaging.

The gradient verifies the ray direction. Differentiating the exact phase gives

$$\frac{1}{n_e} \frac{\partial T_{\text{ideal}}}{\partial r} = -\frac{r}{\sqrt{f^2 + r^2}}, \quad (11.8)$$

which is precisely the transverse direction cosine of a ray aimed from radius r to F . Thus the phase and ray descriptions agree without a paraxial approximation for this ideal scalar screen.

11.4 A quadratic phase lens and its residual aberration

The common paraxial lens phase retains only $T_q = -n_e r^2/(2f)$. Subtract the exact ideal phase of Eq. (11.6):

$$W_q = T_q - T_{\text{ideal}} = -\frac{n_e r^4}{8f^3} + \frac{n_e r^6}{16f^5} + O(r^8). \quad (11.9)$$

A quadratic phase is exactly the paraxial focusing solution, yet it has a residual relative to exact spherical convergence at finite aperture. This provides a useful example of an approximation error appearing as Zernike aberration.

Take $f = 100$ mm, $a = 5$ mm, and $n_e = 1$. The quartic coefficient is $A = -78.125$ nm and the sixth-power coefficient is $B = 0.09765625$ nm. To leading quartic order,

$$c_2^0 = -22.553 \text{ nm}, \quad c_4^0 = -5.823 \text{ nm}. \quad (11.10)$$

Removing piston and paraxial defocus leaves about 5.823 nm RMS. Equation (9.5) predicts a reference-focus shift of -0.0625 mm. This example describes an ideal quadratic phase element; a real spherical singlet with the same focal length need not have these coefficients.

11.5 Relating the thin-lens power to surface curvature

For a thin lens with front and rear signed radii R_1, R_2 , use a forward coordinate z along light propagation and the convention that a positive radius has its centre of curvature to the right. To quadratic order,

$$t(r) - t_0 \simeq \frac{r^2}{2R_2} - \frac{r^2}{2R_1}. \quad (11.11)$$

The lens phase is $(n_g - n_e)[t(r) - t_0]$. Comparing its quadratic term with $-n_e r^2 / (2f)$ gives

$$\boxed{\frac{1}{f} = \left(\frac{n_g}{n_e} - 1\right) \left(\frac{1}{R_1} - \frac{1}{R_2}\right)}. \quad (11.12)$$

For a symmetric biconvex lens in air, $R_1 > 0$ and $R_2 < 0$, so the power is positive. A positive focal length corresponds to convergence. This derivation exposes both the origin and assumptions of the lensmaker equation: it matches quadratic phase, neglects separation between powered surfaces, and uses nearly axial propagation.

It is tempting to expand each sag to fourth power and declare the resulting thickness term to be the exact spherical aberration of a singlet. That step is not justified by the quadratic derivation. Oblique internal propagation and the changed second-surface intercept also contribute at the order of spherical aberration. For quantitative singlet design, use the exact prescription-to-wavefront procedure in the next chapter and then fit the resulting OPD.

11.6 Dispersion: coefficients depend on wavelength

A geometrical thickness map can be fixed while its phase changes with wavelength. For a small error in the thin approximation,

$$c_j(\lambda_0) = [n_g(\lambda_0) - n_e(\lambda_0)]b_j, \quad c_j^{\text{waves}}(\lambda_0) = \frac{c_j(\lambda_0)}{\lambda_0}. \quad (11.13)$$

There are two separate wavelength dependences: the OPD changes through dispersion, and conversion from length to waves divides by wavelength. In a powered system the ideal reference focus and the pupil mapping can also change. To distinguish chromatic focus from residual shape, report whether each wavelength was refocused independently or all wavelengths used one fixed detector plane. Those are different performance questions.

An index inhomogeneity contributes, to first order along the nominal ray, $\delta W = \int \delta n \, ds$. A stress-induced index distribution may also be polarization dependent. Therefore two lenses with the same measured surface shapes need not have identical transmitted wavefronts.

Chapter 12

From an optical prescription to a fitted wavefront

12.1 What a prescription must contain

A ray-traceable prescription specifies each surface's position and orientation, sag or implicit equation, clear aperture, medium on each side, wavelength-dependent refractive index, and the incident rays. It also specifies the stop, object position or field direction, and the output reference. “A lens of focal length 50 mm” is not enough: many different lens shapes share that paraxial focal length and have different aberrations.

Represent a ray by its current point $\mathbf{r}_0$, unit propagation direction $\mathbf{s}$, refractive index n , and accumulated optical path S . In a homogeneous medium its path is

$$\mathbf{r}(\ell) = \mathbf{r}_0 + \ell \mathbf{s}. \quad (12.1)$$

For an implicit surface $F(\mathbf{r}) = 0$, solve $F(\mathbf{r}_0 + \ell \mathbf{s}) = 0$ for the physically relevant intersection. A quadratic sphere can yield two roots. Choose the root belonging to the intended surface branch and forward ray segment, not simply the smallest algebraic number. Check the clear aperture before continuing. The optical path increment is $n\ell$.

For a sag surface $F = z - z_s(x, y)$, an unnormalized normal is $(-\partial_x z_s, -\partial_y z_s, 1)$. Normalize it and orient it consistently. Intersections near tangency require care because a small position error can lead to a large intersection change.

12.2 Vector refraction derived from tangential phase continuity

Let $\mathbf{N}$ point from the second medium toward the first, and require $c_i = -\mathbf{s}_i \cdot \mathbf{N} \geq 0$. Continuity of phase along the interface means continuity of tangential optical momentum:

$$n_i[\mathbf{s}_i - (\mathbf{s}_i \cdot \mathbf{N})\mathbf{N}] = n_o[\mathbf{s}_o - (\mathbf{s}_o \cdot \mathbf{N})\mathbf{N}]. \quad (12.2)$$

Set $\eta = n_i/n_o$. The transmitted tangential direction is therefore $\eta(\mathbf{s}_i + c_i \mathbf{N})$. Its squared magnitude is $\eta^2(1 - c_i^2)$. Since the transmitted unit vector points into the second medium, its normal component is $-c_t \mathbf{N}$, where

$$c_t = \sqrt{1 - \eta^2(1 - c_i^2)}. \quad (12.3)$$

Adding the tangential and normal components gives

$$\boxed{\mathbf{s}_o = \eta \mathbf{s}_i + (\eta c_i - c_t) \mathbf{N}}. \quad (12.4)$$

If the square-root argument is negative, there is total internal reflection and no propagating transmitted ray of this type. At normal incidence, $c_i = c_t = 1$ and $\mathbf{s}_i = -\mathbf{N}$, so the formula returns $\mathbf{s}_o = -\mathbf{N}$, an immediate check. This construction is equivalent to Snell's law but works without choosing a meridional plane.

12.3 Accumulating optical path at common output points

After the last surface, propagate each ray to one output plane before a caustic makes the field multivalued. Let the intersection be $Q_j = (x_j, y_j, z_p)$ and the accumulated actual eikonal be S_j . Choose an ideal focus $F = (x_f, y_f, z_f)$ and output index n' . An ideal converging spherical eikonal on this plane is

$$S_{\text{id}}(Q) = C - n' \|F - Q\|. \quad (12.5)$$

Consequently,

$$\boxed{W_j = S_j + n' \|F - Q_j\| - C.} \quad (12.6)$$

Choose C so that a reference ray has zero OPD, or remove the fitted piston afterwards. The distance in the ideal term goes from the same point Q_j to the same chosen focus for every ray. Adding an actual ray's path to its *own* axial intersection would compare different endpoints and would not define one wavefront error.

For an off-axis image, x_f and y_f need not be zero. For an ideal plane output wave with unit direction $\mathbf{s}_{\text{id}}$, instead use $S_{\text{id}} = C + n' \mathbf{s}_{\text{id}} \cdot \mathbf{Q}$. The chosen ideal wave should match the physical question: a fixed detector, a Gaussian image, a best-fit image, or a collimated output.

The OPD samples are now attached to output coordinates Q_j . If coefficients are to describe an exit pupil, map to that pupil and use its area measure. If one labels the same OPD samples by entrance ray coordinates, that is a legitimate *pullback* representation, but it has a different definition. The Jacobian of the mapping matters for averages: if $Q = Q(u, v)$,

$$dA_{\text{out}} = \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du dv. \quad (12.7)$$

Uniform input-ray sampling need not be uniform in the output pupil.

12.4 An exact finite-thickness singlet example

Consider a symmetric biconvex lens in air with

$$R_1 = +50 \text{ mm}, \quad R_2 = -50 \text{ mm}, \quad t = 5 \text{ mm}, \quad n_g = 1.50. \quad (12.8)$$

The front vertex is at $z = 0$, the rear vertex at $z = 5 \text{ mm}$, and incident rays travel along $+z$. Use a ray vector (y, p) with $p = n\theta$ in paraxial optics. A refracting surface of power $\Phi = (n_o - n_i)/R$ and a translation d in index n have matrices

$$M_{\text{surf}} = \begin{pmatrix} 1 & 0 \\ -\Phi & 1 \end{pmatrix}, \quad M_{\text{trans}} = \begin{pmatrix} 1 & d/n \\ 0 & 1 \end{pmatrix}. \quad (12.9)$$

Both lens surfaces have power 0.01 mm^{-1} in these coordinates. Multiplying in propagation order gives

$$M = M_2 M_t M_1 = \begin{pmatrix} 0.966666667 & 3.333333333 \text{ mm} \\ -0.019666667 \text{ mm}^{-1} & 0.966666667 \end{pmatrix}. \quad (12.10)$$

If its entries are $(A, B; C, D)$, a collimated input ray has output $y = Ay_0$, $\theta = Cy_0$ in air. Its paraxial axis crossing is $-A/C$ after the rear vertex. Thus

$$f_{\text{EFL}} = -\frac{1}{C} = 50.847458 \text{ mm}, \quad f_{\text{BFL}} = -\frac{A}{C} = 49.152542 \text{ mm}. \quad (12.11)$$

The effective focal length and back focal length differ because the principal plane is not at the rear vertex. The Gaussian focus is at $z_f = 54.152542 \text{ mm}$.

Now trace the actual spherical intersections and use Eq. (12.4), without replacing angles by slopes. Compare the actual eikonal on plane $z_p = 6 \text{ mm}$ with the ideal wave converging to this Gaussian focus. Set the axial ray's OPD to zero. The resulting values are:

Input h (mm)	Output y_6 (mm)	Ray axis z (mm)	$W(y_6)$ (μm)
1	0.946929	54.121294	-0.003020
2	1.893426	54.027348	-0.048357
3	2.839043	53.870092	-0.245056
4	3.783299	53.648489	-0.775586
5	4.725660	53.361050	-1.896914

At $h = 5 \text{ mm}$ the ray crosses the axis about 0.7915 mm before the Gaussian focus; its transverse intercept at the Gaussian plane is -0.078975 mm . These are ray aberrations. The last column is a wavefront comparison at a common plane, a different quantity with different units and interpretation.

The supplied numerical companion samples this output OPD and, for one explicit illustration, labels it by the *entrance* coordinate $\rho = h/(5 \text{ mm})$ with uniform entrance-disk weighting. The resulting pullback coefficients through $n = 8$ are approximately

$$c_0^0 = -0.631518947 \mu\text{m}, \quad c_2^0 = -0.547320957 \mu\text{m}, \quad (12.12)$$

$$c_4^0 = -0.141739671 \mu\text{m}, \quad c_6^0 = -0.000177324 \mu\text{m}, \quad (12.13)$$

$$c_8^0 = +0.000000560 \mu\text{m}. \quad (12.14)$$

They are not presented as exit-pupil coefficients. The nonzero defocus coefficient is not an inconsistency with choosing the Gaussian focus: the higher-power wavefront contains a projection onto Z_2^0 at finite aperture. Refitting after moving the ideal focus changes that projection.

12.5 Checks that distinguish a model from a trustworthy calculation

A useful ray tracer should pass checks with known consequences. A plane-parallel plate returns the emerging ray parallel to the incident ray in identical external media. A paraboloid returns zero on-axis reference OPD in the ideal geometry. Rays near the axis approach the paraxial matrix result. The eikonal gradient at a common output plane agrees with actual minus ideal transverse optical momentum:

$$\nabla_{\perp} W = n'(\mathbf{s}_{a,\perp} - \mathbf{s}_{i,\perp}). \quad (12.15)$$

For the numerical singlet, a finite-difference derivative near the ray launched at $h = 3 \text{ mm}$ gives $\partial W/\partial y_6 \simeq -3.4606372 \times 10^{-4}$, while the independent difference of direction cosines gives $-3.4606368 \times 10^{-4}$. Agreement checks signs, path accumulation, and the common-plane mapping simultaneously.

Do not demand that the simple paraxial spot relation $\Delta \mathbf{x} = (L/n')\nabla_{\perp} W$ reproduce every higher-order term of an exact high-aperture ray trace. Exact propagation uses $\mathbf{x}_f = \mathbf{x}_p + L\mathbf{s}_{\perp}/s_z$; the derivative of the factor $1/s_z$ also matters beyond the leading paraxial approximation.

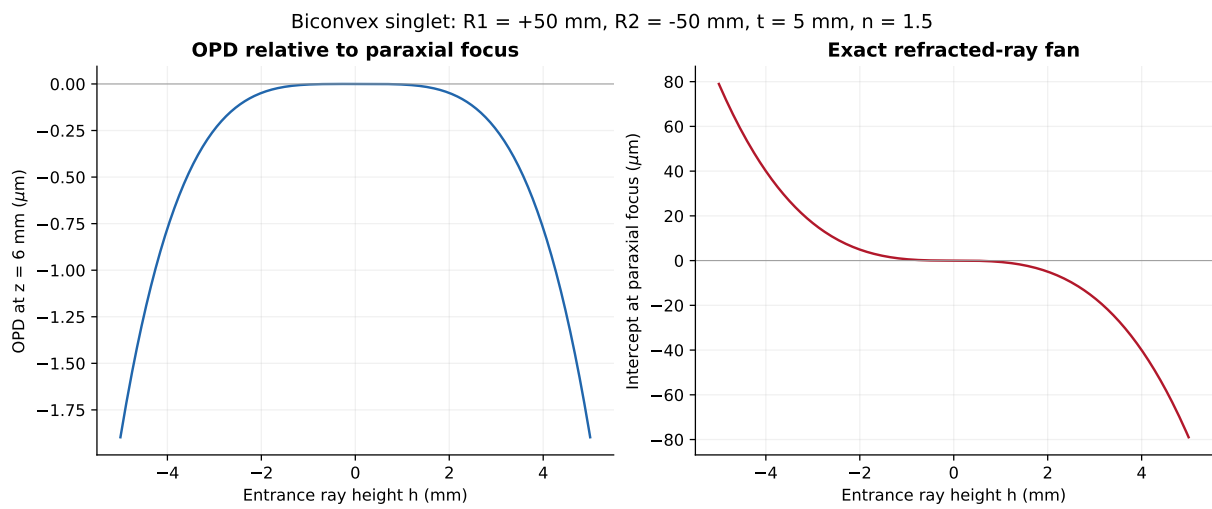


Figure 12.1: Exact ray and OPD checks for the finite singlet. The horizontal coordinate is entrance ray height. The comparison explicitly includes refraction at both surfaces and optical travel inside the glass.

Chapter 13

Fitting measured data without losing its meaning

13.1 From an integral to a matrix

An instrument supplies samples w_i at positions (x_i, y_i) , often with missing values and nonuniform uncertainty. Choose M retained modes and form the design matrix

$$A_{ij} = Z_j(\rho_i, \theta_i), \quad \mathbf{w} \simeq A\mathbf{c}. \quad (13.1)$$

For positive weights q_i , fit by minimizing

$$J(\mathbf{c}) = \sum_i q_i \left[w_i - \sum_j A_{ij} c_j \right]^2 = \|Q^{1/2}(\mathbf{w} - A\mathbf{c})\|_2^2, \quad (13.2)$$

where $Q = \text{diag}(q_i)$. Equal-area pixels can use equal area weights. A polar grid uniform in ρ needs weights proportional to ρ away from the origin, or preferably a quadrature designed for area integration. Independent measurements with known noise variances σ_i^2 instead lead to statistical weights $q_i = 1/\sigma_i^2$ for a Gaussian-noise maximum-likelihood fit.

Area weighting and inverse-variance weighting answer different questions. The former approximates a physical pupil RMS objective; the latter extracts coefficients efficiently under a noise model. They can be combined intentionally, but then the fitting objective must be named. The RMS used to report optical performance can be computed with a separate physical pupil measure after the fit.

Differentiating Eq. (13.2) gives the normal equations

$$A^T Q A \mathbf{c} = A^T Q \mathbf{w}. \quad (13.3)$$

They are helpful for understanding the geometry, but explicitly inverting $A^T Q A$ is a poor default numerical method. Forming this product squares the condition number of the weighted design matrix and can amplify roundoff.

13.2 QR and singular-value decomposition

Set $B = Q^{1/2}A$ and $\mathbf{y} = Q^{1/2}\mathbf{w}$. A thin QR factorization $B = UR$ has orthonormal columns in U and an upper triangular matrix R . For full column rank,

$$R\mathbf{c} = U^T \mathbf{y}. \quad (13.4)$$

Solve this triangular system by back-substitution. There is no need to compute R^{-1} explicitly.

A singular-value decomposition is

$$B = U\Sigma V^T, \quad \Sigma = \text{diag}(s_1, \dots, s_r), \quad s_1 \geq \dots > 0. \quad (13.5)$$

It gives the least-squares solution

$$\mathbf{c} = \sum_{k=1}^r \frac{\mathbf{u}_k^T \mathbf{y}}{s_k} \mathbf{v}_k. \quad (13.6)$$

Each $\mathbf{v}_k$ is a combination of modes. A tiny singular value means that this combination produces very little signal on the measured samples; noise divided by s_k can then make its coefficient huge. Truncating such directions stabilizes the fit but also declares them unresolved. Report rank and the singular-value threshold. The supplied code uses NumPy's documented least-squares interface, which returns the rank and singular values [4].

13.3 Why masked projection is not least squares

On a full, continuously sampled disk our normalized modes obey $\langle Z_j, Z_k \rangle_D = \delta_{jk}$. After a central obstruction or missing sector, their Gram matrix becomes

$$G_{jk} = \langle Z_j, Z_k \rangle_{\text{mask}}, \quad (13.7)$$

and generally $G \neq I$. Taking each masked dot product as a coefficient gives $\mathbf{b} = G\mathbf{c}$ even for perfectly noiseless data in the model. Thus each reported value includes contributions from other modes. Least squares solves the coupled problem instead.

The numerical companion constructs a known combination of 28 modes through $n = 6$ and removes an obscured centre, a support strip, and a sector. Correct weighted least squares recovers the coefficients to about $10^{-16} \mu\text{m}$ in that noiseless numerical example. Treating normalized masked dot products as coefficients produces a maximum error of approximately $0.0205 \mu\text{m}$. This is a controlled demonstration of mode mixing, not a claim that real metrology achieves machine-precision accuracy.

13.4 Residuals, uncertainty, and the cost of adding modes

After fitting, compute $r_i = w_i - (A\hat{\mathbf{c}})_i$. Plot the residual as a map. A single residual RMS cannot reveal whether the remaining error is a scratch, a clipped edge, a periodic tool mark, or a smooth omitted mode. Examine residual structure, sampling scale, and repeated measurements.

If the noise vector has covariance Σ_ϵ and the fit is linear, $\hat{\mathbf{c}} = H\mathbf{w}$ for an estimator matrix H . Hence

$$\text{Cov}(\hat{\mathbf{c}}) = H\Sigma_\epsilon H^T. \quad (13.8)$$

When $Q = \Sigma_\epsilon^{-1}$ and the model has full column rank, the algebraic covariance is

$$\text{Cov}(\hat{\mathbf{c}}) = (A^T \Sigma_\epsilon^{-1} A)^{-1}. \quad (13.9)$$

This identity explains the statistics; evaluate it through a stable factorization, not an unnecessary explicit normal-matrix inverse. For independent noise of unknown common standard deviation, a residual estimate is $\hat{\sigma}^2 = \sum_i r_i^2 / (N - r)$ with fit rank r , provided the model is adequate and the samples are independent. Correlated pixels reduce the effective information; simply counting every pixel as independent understates uncertainty.

For a scalar full-disk RMS $\sigma = \sqrt{\mathbf{c}^\top P \mathbf{c}}$, where P selects the retained error coefficients, first-order error propagation gives

$$\text{Var}(\sigma) \simeq \frac{(P\mathbf{c})^\top \text{Cov}(\mathbf{c})(P\mathbf{c})}{\sigma^2}. \quad (13.10)$$

This linear approximation fails near zero RMS, where the norm is nonlinear and the denominator vanishes. Monte Carlo propagation or an appropriate confidence bound is then more useful.

Adding modes cannot increase the training residual in ordinary least squares, but can fit noise and make coefficients unstable. Choose an order using the spatial resolution, expected physics, singular values, repeatability, and prediction on withheld data. A small change in pupil edge location should not produce a large unsupported change in the research conclusion. Repeat the fit over a reasonable range of orders and report which conclusions are stable.

13.5 Fitting slopes measured by a wavefront sensor

A Shack–Hartmann sensor measures local ray directions through spot displacements. If a lenslet has focal length f_ℓ and output index n' , the local paraxial relation is

$$\Delta x_\ell \simeq \frac{f_\ell}{n'} \langle \partial_x W \rangle_\ell, \quad \Delta y_\ell \simeq \frac{f_\ell}{n'} \langle \partial_y W \rangle_\ell. \quad (13.11)$$

The average is over a lenslet footprint with appropriate illumination weighting. Build a design matrix from averaged derivatives of the Zernikes, stack the two slope components, and solve the same weighted least-squares problem. Point evaluation at lenslet centres is an additional approximation for modes varying appreciably across a lenslet.

Piston has zero derivative and is unobservable from slopes alone; its column is identically zero. Remove it from the fit or impose a gauge such as zero mean. Disconnected pupil regions can have additional unobserved differential pistons. A numerical pseudoinverse selecting zero for an unobserved quantity does not constitute a measurement of that quantity. ESO’s instrument documentation provides an example of wavefront-sensor measurements used to guide mirror corrections [5].

Chapter 14

Changing pupils, coordinates, and illumination

14.1 Rotating a wavefront without changing its strength

Take one angular pair with coefficients (a_c, a_s) multiplying $NR_n^m \cos m\theta$ and $NR_n^m \sin m\theta$, $m > 0$. Physically rotate the pattern counterclockwise by α while holding the axes fixed. The new map is $W_{\text{new}}(\rho, \theta) = W_{\text{old}}(\rho, \theta - \alpha)$. Expand both trigonometric functions:

$$\cos[m(\theta - \alpha)] = \cos m\theta \cos m\alpha + \sin m\theta \sin m\alpha, \quad (14.1)$$

$$\sin[m(\theta - \alpha)] = \sin m\theta \cos m\alpha - \cos m\theta \sin m\alpha. \quad (14.2)$$

Collecting coefficients gives

$$\begin{pmatrix} a'_c \\ a'_s \end{pmatrix} = \begin{pmatrix} \cos m\alpha & -\sin m\alpha \\ \sin m\alpha & \cos m\alpha \end{pmatrix} \begin{pmatrix} a_c \\ a_s \end{pmatrix}. \quad (14.3)$$

The matrix preserves $a_c^2 + a_s^2$. A rotation mixes members of an angular pair, not different radial degrees. A passive rotation of the coordinate axes uses the opposite angle. A reversal $Y \mapsto -Y$ leaves cosine modes unchanged and reverses the signs of sine modes.

For example, pure cosine astigmatism rotated through 45° becomes pure sine astigmatism because $m\alpha = 90^\circ$. Pure cosine trefoil rotated through 30° similarly becomes pure sine trefoil. The factor m is why mode orientation cannot be read by treating every pair like ordinary tilt.

14.2 Changing the pupil radius mixes radial orders

Suppose a measured physical wavefront was fitted on radius a , and we now restrict it to the concentric radius $b = ka$ with $0 < k < 1$. For the new normalized coordinate $\rho' = r/b$, $\rho = k\rho'$. Even though the physical wavefront has not changed, the polynomial arguments and the averaging domain have changed.

For pure defocus,

$$c_2^0 Z_2^0(k\rho') = k^2 c_2^0 Z_2^0(\rho') + \sqrt{3}(k^2 - 1)c_2^0. \quad (14.4)$$

The new defocus coefficient scales by k^2 , and the new pupil mean changes. For pure primary spherical aberration, substitute $k\rho'$ into Eq. (8.12) and use the monomial identities:

$$c_4^{0'} = k^4 c_4^0, \quad (14.5)$$

$$c_2^{0'} = \sqrt{15}(k^4 - k^2)c_4^0, \quad (14.6)$$

$$c_0^{0'} = \sqrt{5}(2k^4 - 3k^2 + 1)c_4^0. \quad (14.7)$$

A pattern balanced on the original pupil is not generally balanced on the smaller one. If the new observation can be refocused, remove the newly generated focus component under the new pupil measure. Scaling a measured coefficient by k^n is therefore valid for the highest radial degree in this example but misses induced lower modes.

Decentring also mixes orders and angular families. Expanding $(X - \delta)^2 + Y^2$ immediately produces a term linear in X ; a decentred focus pattern contains tilt and piston. Rather than relying on a large table of transformation coefficients, evaluate the original polynomials at transformed coordinates and refit on the desired pupil. This method also handles combined translations, rotations, and scales, provided the original map is known over the needed region. Extrapolating beyond a measured aperture is an unsupported model prediction.

14.3 Annular pupils: derive the simplest corrected mode

Let a central obstruction remove $0 \leq \rho < \epsilon$. The normalized annular inner product is

$$\langle f, g \rangle_A = \frac{1}{\pi(1 - \epsilon^2)} \int_0^{2\pi} \int_\epsilon^1 f g \rho \, d\rho \, d\theta. \quad (14.8)$$

The annular mean of ρ^2 is

$$\frac{2}{1 - \epsilon^2} \int_\epsilon^1 \rho^3 \, d\rho = \frac{1 + \epsilon^2}{2}. \quad (14.9)$$

Thus ordinary disk defocus has annular mean $\langle Z_2^0, 1 \rangle_A = \sqrt{3}\epsilon^2$ and is no longer orthogonal to piston. With $u = \rho^2$, the area measure is uniform on $[\epsilon^2, 1]$, whose variance is $(1 - \epsilon^2)^2/12$. Subtracting the mean and normalizing produces

$$\boxed{Z_{2,A}^0 = \frac{\sqrt{3}[2\rho^2 - (1 + \epsilon^2)]}{1 - \epsilon^2}}. \quad (14.10)$$

This derivation illustrates the principle behind annular orthogonal bases. Their coefficients are not interchangeable with coefficients of ordinary disk polynomials restricted to an annulus.

Either representation can be useful. Ordinary disk functions retain familiar shape labels and can be fitted together by least squares. An orthonormal annular or numerically orthogonalized basis gives simple RMS sums for that aperture. State which basis was used and preserve the mask that defines it.

14.4 Elliptical and grazing-incidence footprints

For an ellipse with semiaxes a_x, a_y , use $X = x/a_x, Y = y/a_y$ after aligning the axes. Its interior maps to a unit disk. Since the area Jacobian is the constant $a_x a_y$, disk Zernikes composed with this mapping are orthonormal for uniform physical area on the ellipse. This is a coordinate construction; it does not make the optical system rotationally symmetric. Physical slopes have different factors $1/a_x$ and $1/a_y$, and optical interpretations of “defocus” or “astigmatism” must account for them.

An oblique beam’s footprint on a flat mirror is stretched along the plane of incidence. At grazing incidence, let γ be the angle above the surface, so $i = \pi/2 - \gamma$ and

$$\delta W = -2nh \sin \gamma. \quad (14.11)$$

The smaller OPD sensitivity per unit normal height does not mean that surface accuracy becomes irrelevant. The footprint is long, the mapping is anisotropic, angular errors still affect the image,

and very short wavelengths make small absolute path errors significant. For curved grazing mirrors, incidence and pupil magnification vary across the footprint; use Eq. (10.6) or direct tracing.

14.5 Amplitude, phase, coatings, and short wavelengths

A pupil field is complex:

$$P(x, y; \lambda_0) = A(x, y; \lambda_0) \exp \left[i \frac{2\pi W_{\text{geom}}(x, y)}{\lambda_0} + i\phi_{\text{coat}}(x, y; \lambda_0) \right]. \quad (14.12)$$

The amplitude A includes illumination, obscurations, transmission, and reflectivity. A spatially varying coating phase acts like an additional OPD

$$W_{\text{coat}} = \frac{\lambda_0}{2\pi} \phi_{\text{coat}}, \quad (14.13)$$

after a continuous phase branch and a piston reference have been chosen. A uniform reflection phase contributes piston; a nonuniform one can contribute focus or higher modes. Near a phase jump, unwrapping and amplitude zeros require particular care.

Multilayer reflectors used in EUV and X-ray optics are interference structures; their complex reflected amplitude depends on wavelength, angle, polarization, and layer design. The CXRO multilayer tutorial provides a primary laboratory introduction to that interference mechanism [6]. In such systems, a surface sag fit alone does not determine the image. Combine geometrical path, coating response, and amplitude propagation. A scalar OPD can also be inadequate when polarization components acquire substantially different phases or directions; the appropriate field model then has vector components.

Chapter 15

From Zernike coefficients to images

15.1 Why rays alone cannot predict the point image

Rays describe local propagation directions and are powerful for layout and aberration diagnosis. They do not by themselves describe interference across a finite pupil. Even a zero-aberration mirror has a finite diffraction pattern. A point-spread function (PSF) describes the intensity produced by one ideal point object; the PSF is the bridge from a wavefront map to an image.

For monochromatic, scalar, paraxial propagation near an ideal focal plane, the complex amplitude has the Fourier-integral form

$$U(\xi, \eta) = C \iint_{\mathcal{P}} A(x, y) \exp \left[ik_0 W(x, y) - i \frac{2\pi n'}{\lambda_0 f} (x\xi + y\eta) \right] dx dy, \quad (15.1)$$

where $k_0 = 2\pi/\lambda_0$, f is the relevant pupil-to-image Fourier scale in this simplified model, and C is a propagation constant. The PSF is proportional to $|U|^2$ with a chosen energy normalization. This form follows by expanding propagation distance to quadratic order: the ideal converging pupil phase cancels the pupil-dependent quadratic term, leaving a linear phase in image coordinates and the residual W .

For an aplanatic high-numerical-aperture system, or when polarization matters, Eq. (15.1) must be replaced by the corresponding vector diffraction integral and correct pupil apodization. The coefficient table remains a description of a phase map, but the propagation model must match the physical regime.

15.2 Deriving the small-error Strehl relation

At the ideal image origin, Eq. (15.1) becomes the sum of pupil phasors. For real nonnegative amplitude A , define the coherent amplitude-weighted average

$$\langle f \rangle_A = \frac{\int_{\mathcal{P}} A f dA}{\int_{\mathcal{P}} A dA}. \quad (15.2)$$

Relative to the ideal image with the *same amplitude pupil*, the normalized central intensity is

$$S_{\text{centre}} = \left| \left\langle e^{ik_0 W} \right\rangle_A \right|^2. \quad (15.3)$$

This is evaluated at the specified image point. If tilt or defocus is present, that point may not be the image maximum. State whether a reported Strehl uses a fixed reference, a recentered image, or an optimized focus.

Choose piston so $\langle W \rangle_A = 0$. Expand the exponential:

$$\langle e^{ik_0 W} \rangle_A = 1 - \frac{k_0^2}{2} \langle W^2 \rangle_A - \frac{ik_0^3}{6} \langle W^3 \rangle_A + \dots \quad (15.4)$$

Multiplying by the complex conjugate and retaining second order gives

$$S_{\text{centre}} \simeq 1 - k_0^2 \sigma_{W,A}^2 \quad (15.5)$$

For weak aberrations the leading image degradation depends on phase variance. A frequently useful extension is

$$S \approx \exp[-(2\pi\sigma_W/\lambda_0)^2], \quad (15.6)$$

usually called the extended Maréchal approximation. The exponential matches the second-order expansion and is exact for the central phasor average of a zero-mean Gaussian distribution of phase values under the averaging measure. It is not a universal exact law for deterministic aberrations. In particular, do not apply it blindly to large phase excursions, sparse phase defects, amplitude errors, or strongly obscured and polarization-dependent systems.

For a uniform full disk, σ_W can be obtained from the appropriate Zernike coefficient sum. For nonuniform amplitude, the variance entering Eq. (15.5) is weighted by A , not automatically by intensity A^2 . Geometrical image statistics, detector photon weights, metrology likelihoods, and coherent focal amplitude therefore need not use the same weights.

15.3 A wavelength comparison and a caution about RMS

An OPD of 30 nm RMS corresponds to phase RMS $2\pi(30/633) = 0.298$ radians at 633 nm. The exponential estimate is $S \simeq 0.915$. At 13.5 nm the same OPD is more than two waves RMS, far outside the weak-aberration regime. The numerical exponential would be tiny, but treating that number as a reliable image prediction would be unjustified; calculate the complex pupil and its diffraction pattern directly.

For a target $S \geq 0.8$ within the weak-error approximation,

$$\sigma_W \lesssim \frac{\lambda_0}{2\pi} \sqrt{-\ln 0.8} \simeq 0.0752\lambda_0. \quad (15.7)$$

This estimate is close to the familiar “about a fourteenth of a wavelength RMS” rule. At 13.5 nm it is roughly 1.02 nm OPD RMS, or 0.51 nm normal-incidence surface RMS for a single ideal mirror in vacuum if all other error sources are absent. A system budget must allocate error among optics, alignment, coatings, and time variation; it cannot assign the entire system allowance independently to every element.

Equal RMS does not imply equal PSF structure. Coma displaces energy asymmetrically; astigmatism changes image shape with focus; a localized defect can create a broad halo. The Strehl approximation compresses image quality into one quantity valid under specified conditions. For research, inspect the PSF, encircled energy, and, where relevant, the optical transfer function in addition to RMS.

15.4 Peak-to-valley error and slope error

For pure defocus $W = cZ_2^0$, the centre and edge values are $-\sqrt{3}c$ and $+\sqrt{3}c$, so

$$\text{PV}_{\text{defocus}} = 2\sqrt{3}|c|. \quad (15.8)$$

For pure balanced spherical aberration, $W = c\sqrt{5}(6u^2 - 6u + 1)$ with $u = \rho^2$. The stationary point satisfies $12u - 6 = 0$, hence $u = 1/2$. The polynomial is 1 at both endpoints and $-1/2$ at $u = 1/2$, giving

$$\text{PV}_{\text{spherical}} = \frac{3}{2}\sqrt{5}|c|. \quad (15.9)$$

Both modes have RMS $|c|$, yet their PV-to-RMS ratios differ. For a sum of modes there is no general rule that adds their individual PVs in quadrature. Extremes must be found from the combined map, with sampling fine enough to resolve them.

Slope also gives a different metric. A small high-spatial-frequency error can have a large gradient even when its height RMS is modest. For a one-dimensional illustrative ripple $W = b\cos(k_x x)$, its height amplitude is b , whereas its slope amplitude is $|bk_x|$. This is why an optical surface specification often needs both low-order figure information and a spatial-frequency or slope description. Low-order Zernike coefficients alone do not constrain roughness or all scattered light.

15.5 Using a Fourier transform in practice

Build a uniformly sampled Cartesian pupil, set the complex amplitude to zero outside the physical mask, evaluate the phase, and transform the complex field. Zero padding interpolates the sampled image more finely; it does not add pupil information or repair an undersampled wavefront. Check that doubling the pupil sampling no longer changes the observables of interest appreciably.

For a grid pitch Δx and transform size N , the spatial-frequency pitch is $\Delta\nu = 1/(N\Delta x)$. In the paraxial model of Eq. (15.1), the image-plane sampling is

$$\Delta\xi = \frac{\lambda_0 f}{n' N \Delta x}. \quad (15.10)$$

Keep the ideal and aberrated pupils on the same grid with the same amplitude and total energy when comparing peaks. If one plot is normalized to its own maximum, its apparent peak no longer shows Strehl. For broadband incoherent imaging, combine intensities at the proper wavelength-dependent image scales; do not add monochromatic complex amplitudes unless the physical fields are mutually coherent.

Chapter 16

A reproducible computational laboratory

16.1 What to calculate before fitting an unknown optic

Before interpreting experimental data, fit a map whose answer is known. This separates questions about your implementation from questions about the optic. The following complete Python example uses only NumPy and the standard library. Coefficients and OPD values are in nanometres. The radial formula is suitable for the modest degree used here; it is not intended as a high-order production evaluator.

The quadrature samples $u = \rho^2$ by Gauss–Legendre integration on $[0, 1]$ and samples angle uniformly. Since $\langle f \rangle_D = \int_0^1 \langle f(\sqrt{u}, \theta) \rangle_\theta du$, the weights sum to one. The code verifies full-disk orthonormality, then fits a deliberately masked pupil. It retains the ordinary disk polynomials and uses least squares for their coupled masked fit.

Listing 16.1: A self-contained Zernike fitting experiment.

```
from math import factorial
import numpy as np
from numpy.polynomial.legendre import leggauss

def zernike(n, m, rho, theta):
    ell = abs(m)
    if n < 0 or ell > n or (n - ell) % 2:
        raise ValueError("Invalid (n, m)")
    rho = np.asarray(rho, dtype=float)
    radial = np.zeros_like(rho)
    for s in range((n - ell) // 2 + 1):
        factor = ((-1)**s * factorial(n - s) /
                  (factorial(s) *
                   factorial((n + ell)//2 - s) *
                   factorial((n - ell)//2 - s)))
        radial += factor * rho**(n - 2*s)
    if m == 0:
        return np.sqrt(n + 1) * radial
    angle = np.cos(m*theta) if m > 0 else np.sin(ell*theta)
    return np.sqrt(2*(n + 1)) * radial * angle

def disk_grid(nr=32, nt=96):
    u, wu = leggauss(nr)
    rho, theta = np.meshgrid(np.sqrt((u + 1)/2),
                              2*np.pi*np.arange(nt)/nt,
                              indexing="ij")
    q = np.broadcast_to(wu[:, None]/(2*nt), rho.shape)
```

```

    return rho.ravel(), theta.ravel(), q.ravel()

def fit(A, w, q):
    sw = np.sqrt(q)
    B = A * sw[:, None]
    c, _, rank, singular = np.linalg.lstsq(B, w*sw,
                                           rcond=None)

    residual = w - A @ c
    rms = np.sqrt(np.sum(q*residual**2)/np.sum(q))
    condition = singular[0]/singular[-1]
    return c, residual, rms, rank, condition

rho, theta, q = disk_grid()
modes = [(n, m) for n in range(7)
          for m in range(-n, n+1, 2)]
A = np.column_stack([zernike(n, m, rho, theta)
                      for n, m in modes])
gram = A.T @ (q[:, None] * A)
assert np.max(np.abs(gram - np.eye(len(modes)))) < 1e-11

truth = np.zeros(len(modes))
specified = {(0, 0): 20., (1, 1): -35., (2, 0): 80.,
             (2, -2): 42., (3, 1): -25., (4, 0): 60.,
             (6, 0): 8.}
for pair, value in specified.items():
    truth[modes.index(pair)] = value
w = A @ truth

x, y = rho*np.cos(theta), rho*np.sin(theta)
valid = ((rho > 0.25) & (np.abs(x) > 0.04)
         & ~((x > 0.3) & (y > 0.25)))
Am, wm, qm = A[valid], w[valid], q[valid]
c, residual, rms, rank, condition = fit(Am, wm, qm)
assert rank == len(modes)
assert np.max(np.abs(c - truth)) < 1e-8
print("rank, condition, residual RMS [nm]:",
      rank, condition, rms)

# This next expression is deliberately wrong for the masked pupil.
naive = Am.T @ (qm*wm) / qm.sum()
print("wrong masked projection error [nm]:",
      np.max(np.abs(naive - truth)))

# An optional noise experiment; area weights stay area weights.
rng = np.random.default_rng(7)
noisy = wm + rng.normal(0., 2., size=wm.size)
cn, rn, rmsn, rankn, condn = fit(Am, noisy, qm)
print("noisy fit residual RMS [nm]:", rmsn)
for pair in specified:
    j = modes.index(pair)
    print(pair, "true, recovered [nm]:", truth[j], cn[j])

```

The random seed makes the final experiment repeatable for a given numerical implementation. Its 2 nm sample noise is an invented teaching input, not an instrument specification. The area weights approximate the masked pupil norm; they are not being claimed as an optimal likelihood weighting for that synthetic noise model. Notice that the printed rank is checked independently of a small residual. An underdetermined fit can also have zero residual.

16.2 Extending the experiment deliberately

First change the known coefficient of one mode and verify that the recovered value follows it. Then remove a larger pupil sector and watch the condition number and coefficient noise sensitivity. Next add a mode of degree higher than the fit includes; the residual should show spatial structure. Finally perturb the assumed pupil centre and radius to observe how geometry errors redistribute coefficients.

To connect the code with a real map, replace the synthetic w and sample coordinates by calibrated measurements, keep explicit units, and build the physical mask. Preserve the raw measurement and its original metadata. Phase in radians must be converted by $W = \lambda_0 \phi / (2\pi)$ only after any necessary phase unwrapping; a wrapped phase map does not describe a continuous OPD by direct pointwise multiplication alone.

The file `../verify_optics.py` supplied alongside these books contains additional symbolic checks, the exact spherical mirror trace, the finite-thickness singlet trace, and figure generation. Its output is recorded in `../numerical_checks.txt`. The code distinguishes surface-footprint, entrance-labelled, and output-plane quantities in its comments and report. Use those distinctions when reproducing a coefficient table.

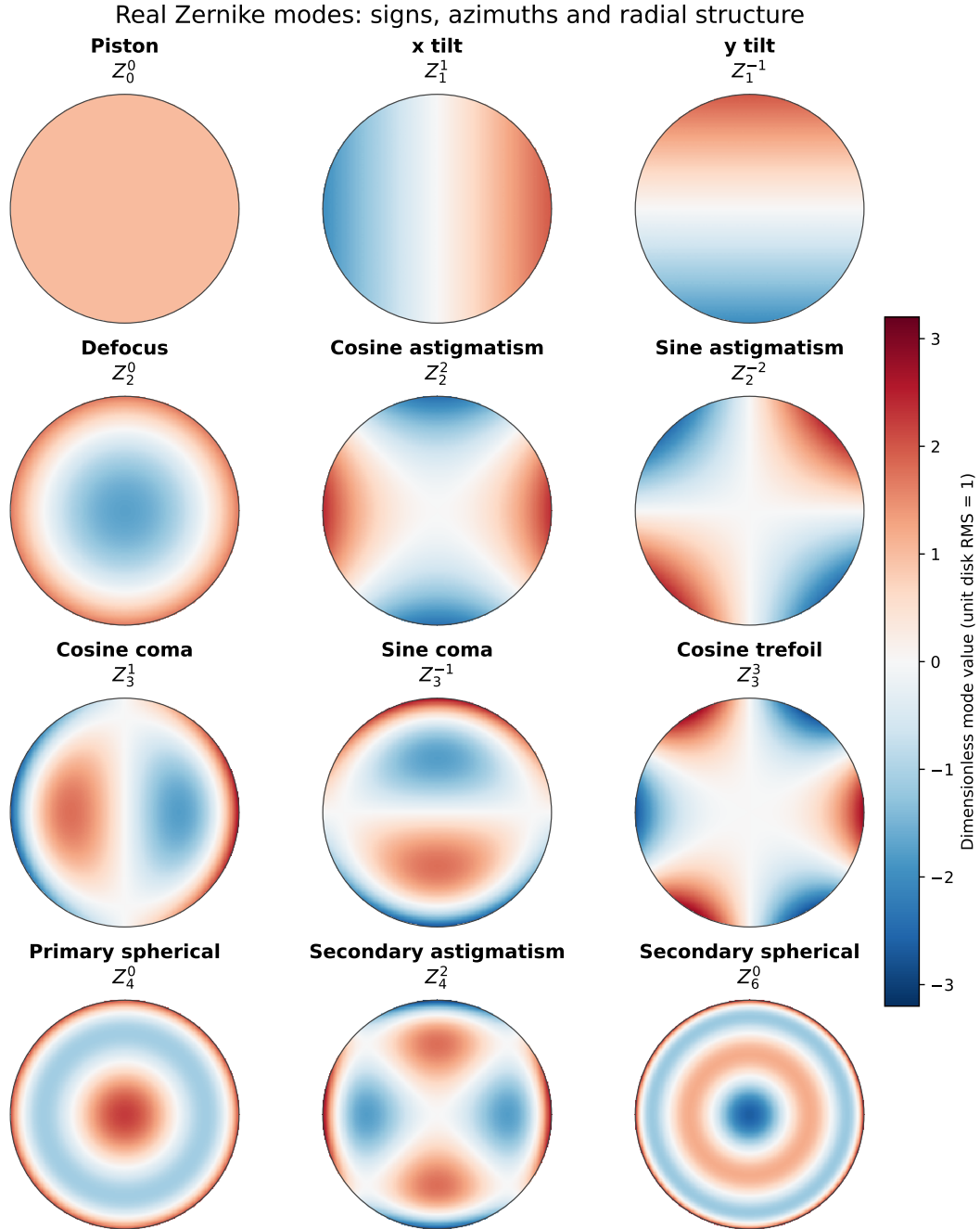


Figure 16.1: Selected real disk-RMS-normalized modes. Colour represents polynomial value, not an assigned physical height. A coefficient with length units supplies that scale. The mode formula and colour scale must both be read when comparing patterns.

Chapter 17

Planning a first research investigation

17.1 A question that can be answered with coefficients

A productive starting question is concrete: “For a specified mirror aperture, object direction, wavelength, and image plane, how much does a particular surface change reduce residual OPD and improve the image?” The answer requires a nominal model, a change, a comparison procedure, and a performance metric. A list of small coefficients by itself does not show that an optical system satisfies its purpose.

Begin with a nominal prescription and an analytic limiting case. For an on-axis paraboloid, verify equal path; for a singlet, verify the paraxial focal lengths. Record the common reference surface and pupil map. Increase aperture or field gradually so you can see when the leading-order formulas stop matching the exact trace. This process turns aberration theory into a controlled approximation whose error can be measured.

17.2 Linear sensitivity and design changes

Let p_k be adjustable prescription parameters: curvature, conic constant, an aspheric coefficient, decentre, or an actuator command. At a nominal design compute coefficients $\mathbf{c}(\mathbf{p})$. For small parameter changes,

$$\mathbf{c}(\mathbf{p} + \delta\mathbf{p}) \simeq \mathbf{c}(\mathbf{p}) + J\delta\mathbf{p}, \quad J_{jk} = \frac{\partial c_j}{\partial p_k}. \quad (17.1)$$

A symmetric finite difference estimates a column:

$$J_{jk} \simeq \frac{c_j(\mathbf{p} + \Delta p_k \mathbf{e}_k) - c_j(\mathbf{p} - \Delta p_k \mathbf{e}_k)}{2\Delta p_k}. \quad (17.2)$$

Choose a step small enough to describe local response but large enough to exceed numerical noise; check the result when the step is halved or doubled. Keep pupil, field, reference, and removed-term policy consistent while taking these differences.

One local correction solves $J\delta\mathbf{p} \simeq -\mathbf{c}$ for the selected error coefficients. Use a weighted and possibly regularized least-squares solve. For example,

$$\min_{\delta\mathbf{p}} \|L(\mathbf{c} + J\delta\mathbf{p})\|_2^2 + \alpha \|R\delta\mathbf{p}\|_2^2. \quad (17.3)$$

Here L scales optical priorities, R scales the cost of changes, and α controls that tradeoff. Bounds can enforce manufacturing or actuator limits. Afterwards, recompute the full nonlinear optical

model. A small residual in the linear model is a proposed step, not proof of the final physical performance.

If two parameters have nearly identical sensitivity columns, their individual values cannot be determined reliably from those coefficients alone. Add another field, wavelength, measurement geometry, or independent surface measurement to distinguish them. This is an identifiability problem, not a failure that can be repaired merely by printing more digits.

17.3 A published case: the Hubble primary-mirror test

NASA's 1990 optical-failure investigation described a roughly 1.3 mm spacing error in a lens of the reflective null corrector used to test the Hubble primary mirror. The report estimated about 0.4 wave RMS of third-order spherical wavefront aberration at 632.8 nm from the as-built mirror information [7]. A null corrector supplies the intended comparison wave for an aspheric test; an error in that comparison can make a precisely manufactured wrong shape appear to pass the test.

Convert the stated wavefront RMS to length and phase:

$$\sigma_W = 0.4(632.8 \text{ nm}) = 253.12 \text{ nm}, \quad (17.4)$$

$$\sigma_\phi = 2\pi(0.4) = 2.5133 \text{ rad}. \quad (17.5)$$

This phase RMS is not small. Therefore the exponential small-error Strehl approximation is not a reliable way to reconstruct the actual telescope image from this one number. Neither can we simply declare $c_4^0 = 253.12 \text{ nm}$ in this book's convention: the historical test, telescope pupil, reference, and normalization must first be matched to our full-disk definition.

The methodological lesson for a new mirror project is to test the reference as carefully as the optic. A self-consistent measurement against one reference can preserve that reference's systematic error. An independent test geometry or independently calibrated reference helps distinguish true surface error from an error in the metrology model. Numerical recovery of synthetic coefficients checks software; it does not validate the physical reference used by an experiment.

17.4 What belongs in a research notebook

For each coefficient table, save the physical prescription or measured map, wavelength and index data, object field, stop and pupil definition, coordinate axes, unit conversions, actual-minus-ideal sign, reference focus, mask, weighting, fitted mode list, and removed terms. Save the residual map, fit rank, and convergence checks beside the table. Include a reconstructed map and at least one image or ray diagnostic whose coordinate scales are labelled.

Compare alternative designs at the same physical requirement. Allowing one design to refocus at every wavelength while forcing another onto a common detector biases the comparison. Similarly, excluding a bad edge region changes the aperture and throughput as well as the RMS. Such choices can be legitimate, but they belong in the experimental question and must appear in the reported result.

For a first study, a useful deliverable is a short reproducible report with an analytic leading-order estimate, an exact ray trace, an OPD fit, a residual map, a diffraction calculation where applicable, and a sensitivity study. The goal is not to make every coefficient zero. It is to understand which physical changes improve the required performance, how robust that improvement is, and what the model still leaves unresolved.

Chapter 18

Exercises with detailed worked solutions

18.1 How to use the problems

Attempt each problem before reading its solution. Keep a units column in numerical work and write the pupil and reference before every coefficient calculation. The problems progress from integral practice to design interpretation. Research extensions deliberately admit more than one sensible modelling choice, but a justified choice is part of the answer.

Problem 1: area averages and an incorrect radial average

Find the disk mean and piston-removed RMS of $W = B\rho^2$. Explain why sampling 100 equally spaced radii with equal weights gives a different answer even as the number of radial samples grows.

Solution. Using $\langle \rho^2 \rangle_D = 1/2$ and $\langle \rho^4 \rangle_D = 1/3$,

$$\overline{W} = \frac{B}{2}, \quad \sigma^2 = B^2 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{B^2}{12}. \quad (18.1)$$

Thus $\sigma = |B|/(2\sqrt{3})$, consistent with the defocus coefficient. Equal radial weights approach an integral in $d\rho$, not in $2\rho d\rho$, and return mean $B/3$. To represent area, sample $u = \rho^2$ uniformly or apply appropriate area weights. Increasing the number of samples does not repair the wrong measure.

Problem 2: build a secondary radial polynomial

Construct R_4^2 using $\rho^4 - \beta\rho^2$, with orthogonality to ρ^2 under the radial measure $\rho d\rho$ and boundary value $R_4^2(1) = 1$.

Solution. The orthogonality condition is

$$\int_0^1 (\rho^4 - \beta\rho^2)\rho^2 \rho d\rho = \frac{1}{8} - \frac{\beta}{6} = 0, \quad (18.2)$$

so $\beta = 3/4$. At the edge the unscaled polynomial is $1/4$; multiply by four to obtain $R_4^2 = 4\rho^4 - 3\rho^2$. Multiplication by $\sqrt{10} \cos 2\theta$ or $\sqrt{10} \sin 2\theta$ gives the two RMS-normalized real modes.

Problem 3: recover coefficients from a Cartesian map

On a unit disk, let $W = (10 + 12X - 8Y + 6X^2 - 6Y^2)$ nm. Find all nonzero coefficients and the piston-removed RMS.

Solution. Use $Z_0^0 = 1$, $Z_1^1 = 2X$, $Z_1^{-1} = 2Y$, and $Z_2^2 = \sqrt{6}(X^2 - Y^2)$. The coefficients are

$$c_0^0 = 10, \quad c_1^1 = 6, \quad c_1^{-1} = -4, \quad c_2^2 = \frac{6}{\sqrt{6}} = \sqrt{6}, \quad (18.3)$$

all in nanometres. The RMS after piston removal is $\sqrt{36 + 16 + 6} = \sqrt{58} = 7.616$ nm. If pointing is also removed, only astigmatism remains and the RMS is $\sqrt{6} = 2.449$ nm. Both answers are valid for different removed-term policies.

Problem 4: balance quartic and sixth-power error

Starting from $W = A\rho^4 + B\rho^6$, find the added polynomial $D\rho^2 + P$ that makes both the disk mean and the defocus coefficient zero. Derive the remaining RMS.

Solution. Equation (9.15) gives $c_2^0 = (D + A)/(2\sqrt{3}) + 9B/(20\sqrt{3})$. Setting this to zero yields $D = -A - 9B/10$. The mean is $P + D/2 + A/3 + B/4$, so

$$P = \frac{A}{6} + \frac{B}{5}. \quad (18.4)$$

The remaining two coefficients are unchanged by piston and defocus: $c_4^0 = (A/6 + B/4)/\sqrt{5}$ and $c_6^0 = B/(20\sqrt{7})$. Hence

$$\sigma_{\text{balanced}} = \sqrt{\frac{(A/6 + B/4)^2}{5} + \frac{B^2}{2800}}. \quad (18.5)$$

Notice that B changes the best defocus and the primary spherical coefficient. It cannot be treated as an isolated secondary spherical mode unless those lower-order changes are compensated.

Problem 5: mirror height and correction sign

A mirror in air has positive normal toward the incident beam. Its measured reflected OPD includes $c_3^1 = +60$ nm. Under normal-incidence, small-deformation assumptions, what surface coefficient produced it, and what added surface coefficient removes it?

Solution. Since $c_3^1 = -2b_3^1$, the inferred departure is $b_3^1 = -30$ nm. The correction must add +30 nm in the same normalized surface mode. It creates -60 nm of OPD and cancels the measured term. A physical actuator solution must still account for influence functions and the actual pupil mapping.

Problem 6: compare normal and grazing incidence

A normal surface-height error is 10 nm in vacuum. Find its OPD magnitude at normal incidence, at $i = 60^\circ$ from the normal, and at a grazing angle $\gamma = 2^\circ$ above the surface.

Solution. Use $|\delta W| = 2|h|\cos i = 2|h|\sin \gamma$ as appropriate. The three magnitudes are 20 nm, 10 nm, and $20 \sin 2^\circ = 0.6980$ nm. The signed values are negative for positive height in our convention. At short wavelength, even the last OPD can be significant. These are local normal-height sensitivities, not complete imaging predictions.

Problem 7: infer a lens thickness coefficient

A weak element in air has index 1.60 and induced OPD coefficient $c_2^{-2} = 18$ nm. Find the corresponding thickness coefficient. If the index becomes 1.58 at a second wavelength, what is that coefficient in OPD length units there?

Solution. The thickness coefficient is $b = 18/0.60 = 30$ nm. At the second wavelength the OPD coefficient is $0.58(30) = 17.4$ nm. To quote it in waves, divide by the second vacuum wavelength in nanometres. The index change alone is insufficient to determine the number of waves.

Problem 8: translate a focus coefficient into a distance

At a pupil of radius $a = 10$ mm in air, the reference focus is $L = 200$ mm downstream and the OPD defocus coefficient is $c_2^0 = +50$ nm. Estimate the reference-focus shift needed to remove this coefficient.

Solution. Convert 50 nm $= 5.0 \times 10^{-5}$ mm. Equation (9.5) gives

$$\delta L = \frac{4\sqrt{3}(200)^2}{(10)^2} (5.0 \times 10^{-5}) \text{ mm} = 0.13856 \text{ mm}. \quad (18.6)$$

The new reference lies farther downstream. Holding the actual wavefront fixed while increasing L increases the ideal quadratic eikonal, thereby decreasing the positive residual focus coefficient.

Problem 9: interpret a rotated astigmatism pair

A map has $c_2^2 = 30$ nm and $c_2^{-2} = 40$ nm. Find its pair RMS and orientation. Then physically rotate the map by 15° and find the new pair.

Solution. The pair RMS is 50 nm and $\psi = \frac{1}{2} \text{atan2}(40, 30) = 26.565^\circ$ modulo 180° . The coefficient rotation angle is $2(15^\circ) = 30^\circ$:

$$a'_c = 30 \cos 30^\circ - 40 \sin 30^\circ = 5.981 \text{ nm}, \quad (18.7)$$

$$a'_s = 30 \sin 30^\circ + 40 \cos 30^\circ = 49.641 \text{ nm}. \quad (18.8)$$

The new pair still has norm 50 nm and its orientation has increased to 41.565° .

Problem 10: stop down a balanced spherical mode

A full pupil has only $c_4^0 = 100$ nm. Restrict the radius by $k = 1/2$. Find the new non-piston coefficients before and after refocusing.

Solution. Equation (14.7) gives $c_4^{0'} = 100/16 = 6.25$ nm and

$$c_2^{0'} = 100\sqrt{15} \left(\frac{1}{16} - \frac{1}{4} \right) = -72.618 \text{ nm}. \quad (18.9)$$

The piston also changes but is not needed for the requested RMS. Before refocusing the error is dominated by the new defocus component. After refocusing the remaining RMS is 6.25 nm in this idealized case. Stopping down reduces the highest-power error strongly, but the original full-pupil balance must be reconsidered.

Problem 11: an annulus makes a familiar mode nonzero-mean

For central obstruction ratio $\epsilon = 0.30$, calculate the annular mean of ordinary disk defocus Z_2^0 . Give an annular unit-RMS defocus mode.

Solution. The mean is $\sqrt{3}\epsilon^2 = 0.155885$, which is nonzero. Using Eq. (14.10),

$$Z_{2,A}^0 = \frac{\sqrt{3}(2\rho^2 - 1.09)}{0.91}, \quad 0.30 \leq \rho \leq 1. \quad (18.10)$$

It has zero annular mean and unit annular RMS. Its normalization depends on the obstruction ratio; changing the obstruction changes the basis.

Problem 12: distinguish an index from a coefficient

One program exports $b = 0.12 \mu\text{m}$ multiplying $F = 6\rho^4 - 6\rho^2 + 1$. Another uses this book's normalized Z_4^0 . Find the second coefficient and the piston-removed RMS.

Solution. Because $Z_4^0 = \sqrt{5}F$, $bF = (b/\sqrt{5})Z_4^0$. The coefficient is $c_4^0 = 0.12/\sqrt{5} = 0.053666 \mu\text{m}$. For this isolated normalized non-piston mode, its RMS is the same magnitude. Merely copying 0.12 into the other program would multiply the represented wavefront by $\sqrt{5}$.

Problem 13: identify an unobservable quantity

You measure exact x and y wavefront slopes everywhere on a connected pupil. Can you determine piston? What changes if the pupil has two disconnected illuminated regions and no relative-phase measurement?

Solution. Adding a constant to W leaves both derivatives unchanged, so absolute piston is unobservable. On two disconnected regions, an independent constant can be added to each while preserving all measured slopes. Their difference is differential piston and affects interference when the fields combine, but the stated measurements do not determine it. Any fit that reports a unique value has imposed an additional constraint or prior; it has not extracted that value from the slopes alone.

Problem 14: derive a leading spherical-mirror ray error

For the mirror at Gaussian focus $f = R/2$, use the leading plane OPD $W(x) = -x^4/(4R^3)$ to estimate the transverse ray error in air. Explain why the same operation on the sixth-order OPD is not an exact fifth-order ray calculation.

Solution. The gradient is $dW/dx = -x^3/R^3$. The paraxial relation gives

$$\Delta x \simeq f \frac{dW}{dx} = -\frac{x^3}{2R^2}. \quad (18.11)$$

It is cubic in pupil coordinate: the primary wave error is quartic, while the corresponding ray error is third order. At higher accuracy, ray propagation involves s_x/s_z , not just s_x . The ideal ray is already inclined away from the axis, so linearizing this ratio introduces additional powers. Reference-plane and pupil-coordinate mapping must also be retained consistently.

Problem 15: a small-error image estimate

A uniformly illuminated circular pupil has nonzero retained error coefficients 20, 15, -10 nm after allowed pointing and focus adjustments. At $\lambda_0 = 550$ nm estimate the RMS and Strehl, and state what is missing from that estimate.

Solution. The RMS is $\sqrt{400 + 225 + 100} = 26.926$ nm. The phase RMS is $2\pi(26.926/550) = 0.3076$ radians, so $S \approx \exp(-0.3076^2) = 0.910$. This omits any unmodelled residual, amplitude variation, polarization effects, and mismatch between the simplified propagation and the actual system. It also does not describe how energy is distributed around the peak. Calculate the PSF if those details matter.

Problem 16: distinguish two planes in a mirror calculation

Two calculations of the same sphere return sixth-power coefficients $1/(8R^5)$ and $9/(8R^5)$ in OPD length per sixth power of radius. Both use the same actual-minus-ideal sign. Is one necessarily wrong?

Solution. No. The first can describe the common-surface optical path function labelled by mirror footprint radius r , and the second the eikonal difference on the virtual vertex plane labelled by intercept x . The two points and their phase propagation differ, and $x = r + r^3/R^2 + \dots$. Require the comparison surface, coordinate map, and reference eikonal before comparing the series. The leading quartic agreement alone is not sufficient to identify the higher-order convention.

Problem 17: test whether fitting more modes helps

You fit experimental data through radial degrees 4, 6, and 8. Training RMS decreases at every step, but the degree-8 coefficients change greatly when one small pupil sector is removed. What should you do?

Solution. Inspect singular values, parameter covariance or repeated-data variability, and the spatial residual. The highest-order combinations may be weakly constrained or absorbing noise and a mask-edge error. Use held-out data or independent repeated maps to compare predictive quality. Check pupil registration and sampling before increasing order again. Retain an order justified by resolution and stability, or use explicit regularization and report it. The training residual alone does not establish that the extra fitted structure is physical.

Problem 18: a first independent design project

Design a computational study comparing a spherical and paraboloidal mirror with the same vertex radius, aperture, axial plane-wave input, and available focus adjustment. Specify the quantities you will plot and the checks that would make the comparison credible.

Worked research outline. Start from $z_s = R - \sqrt{R^2 - r^2}$ and $z_p = r^2/(2R)$. Verify exact constant path for the paraboloid and the quartic leading term for the sphere. Trace reflected rays and calculate common-plane OPD for both, preserving the ray-to-pupil map. Plot axial ray crossings, transverse intercepts, fixed-focus OPD, balanced OPD, and radial coefficients against aperture ratio a/R .

Find best phase-variance focus by an actual one-parameter search of the reference location, and compare it with the paraxial prediction. For a chosen wavelength, compute diffraction images on the same normalized energy scale and compare peak, encircled energy, and shape. Repeat with finer ray or pupil sampling and a higher fit order until the reported metrics converge. Then add a small field angle and observe that the paraboloid's on-axis advantage does not eliminate off-axis aberrations. A useful report explains where each analytic approximation works, where it fails, and how the result depends on allowed refocusing and aperture.

Notation and formula reference

Symbol or expression	Meaning and condition
a	Physical radius of the pupil used to normalize coordinates.
ρ, θ	Dimensionless pupil radius and counterclockwise polar angle.
z, h	Surface sag and departure; positive direction must be stated.
S, W	Eikonal in optical-length units, and actual-minus-ideal eikonal.
λ_0, k_0	Vacuum wavelength and $2\pi/\lambda_0$.
n, m in Z_n^m	Radial degree and signed angular label; these are distinct from refractive index, also often written n .
$m > 0, m < 0$	Cosine and sine respectively in this book's real basis.
c_n^m, b_n^m	OPD coefficients and surface or thickness coefficients, respectively; all carry length units unless stated in waves.
$c_n^m = \langle W, Z_n^m \rangle_D$	Projection on a full uniform disk; masked data generally require a coupled fit.
$\sigma_W^2 = \sum' c_j^2$	Full-disk orthonormal RMS, with the prime excluding piston and any other explicitly removed modes; add unfitted residual power.
$\delta W = -2nh \cos i$	First-order mirror sensitivity for positive normal height toward the incident medium.
$\delta W = (n_g - n_e)\delta t$	Thin, nearly axial transmitting thickness error replacing external medium.
$\nabla_\perp W/n'$	Actual-minus-ideal transverse direction cosine on a common output plane.
$\Delta \mathbf{x} \simeq (L/n')\nabla_\perp W$	Forward paraxial transverse ray displacement at propagation distance L .
$S \approx e^{-(2\pi\sigma_W/\lambda_0)^2}$	Weak-aberration image estimate with the stated pupil and image-reference conditions.

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